

Psychometric Assessment Architecture and Construct Validation Report for Secondary School Educational Guidance in Andhra Pradesh and Telangana

Executive Summary

The transition from secondary education to higher secondary studies in the Indian states of Andhra Pradesh and Telangana represents an educational bottleneck governed by high academic stakes, structural tracking rigidity, and severe information asymmetry¹. Upon completing Class 10 (Secondary School Certificate or equivalent), adolescents typically aged 15 to 17 are required by the Board of Intermediate Education, Andhra Pradesh (BIEAP) and the Telangana State Board of Intermediate Education (TSBIE) to enroll in distinct, specialized four-subject academic combinations³. The overwhelming majority of enrollments concentrate within four tracks: Mathematics, Physics, Chemistry (MPC); Biology, Physics, Chemistry (BiPC); Mathematics, Economics, Commerce (MEC); and Commerce, Economics, Civics (CEC)¹. In practice, this high-stakes decision is heavily distorted by localized occupational prestige hierarchies, aggressive commercial coaching marketing, and intense familial pressure, driving students toward engineering and medical tracks regardless of intrinsic interest or foundational readiness⁵.

The assessment architecture developed for Naaguru addresses this structural misalignment. The platform does not function as a high-stakes screening exam, a clinical diagnostic battery, or a deterministic system for prescribing lifetime careers⁵. Rather, it delivers a mobile-first, bilingual (English and Telugu) psychometric exploration tool designed to evaluate student vocational interests, learning task preferences, and domain-specific self-efficacy⁹. This evaluation translates self-reported preferences into non-deterministic, exploratory Intermediate stream and career cluster recommendations⁵.

Grounding this instrument in contemporary vocational psychology requires critically evaluating traditional Western psychometric models—predominantly Holland's RIASEC circumplex—and adapting them to the socio-developmental realities of South India⁵. Structural meta-analyses demonstrate that Holland's circular order assumptions frequently deteriorate in non-Western, developing economic contexts due to structural labor market barriers, collectivist family decision-making, and strong occupational prestige hierarchies⁷. Synthesizing Holland's person-environment fit principles with Gideon Arulmani's Cultural Preparation Process Model indicates that an adolescent guidance instrument in India must assess task-level enjoyment, perceived learning ease, and real-world educational affinities rather than abstract occupational titles⁵.

The proposed assessment engine uses a 40-item operational battery (36 scored vocational

interest items and 4 unscored contextual items) selected from a 64-item candidate pool. The psychometric framework measures six non-redundant, culturally adapted dimensions:

- Investigative & Scientific Inquiry (ISI)
- Quantitative & Computational Reasoning (QCR)
- Technical & Mechanical Design (TMD)
- Commercial & Economic Enterprise (CEE)
- Social, Helping & Administrative Coordination (SHC)
- Creative & Expressive Arts (CEA)

The instrument adheres strictly to the International Test Commission (ITC) Guidelines for Translating and Adapting Tests, ensuring functional, semantic, and measurement equivalence across English and Telugu¹⁰. Readability is pegged to a Flesch-Kincaid Grade Level of 4.0 to 5.5 to ensure comprehension across urban, semi-urban, and rural student populations¹⁷. The resulting reporting framework replaces deterministic labels with probabilistic, strength-based guidance that fosters constructive family dialogue and informed academic exploration⁵.

What Naaguru Should Actually Measure

A primary vulnerability in educational technology guidance tools is the conflation of subjective interest with objective cognitive aptitude, and the over-interpretation of personality traits from brief self-report inventories⁸. To maintain psychometric credibility and avoid misleading students at a critical academic crossroad, Naaguru V1 must establish strict construct boundaries⁸.

Target Construct Category	In-Scope Operationalization for Naaguru V1	Explicitly Out of Scope for Naaguru V1	Justification & Psychometric Boundaries
Vocational Interests	Affective preference, task enjoyment, and intrinsic curiosity toward specific learning tasks, academic disciplines, and foundational work environments ⁹ .	Passive lifestyle leisure habits with low occupational transferability (e.g., social media browsing, casual gaming).	Meta-analytic research demonstrates that vocational interests show substantial longitudinal stability from mid-adolescence onward ($r =$), directly predicting academic persistence and educational satisfaction ⁹ .

Cognitive Aptitude / Ability	Self-reported relative ease and subjective academic comfort across secondary subjects (treated strictly as perceived affinity, not verified maximum capacity) ⁸ .	Standardized cognitive aptitude testing, psychometric intelligence quotients (<i>g</i>), timed numerical/spatial problem-solving ⁸ .	Measuring true cognitive aptitude requires high-stakes, timed, standardized testing batteries resistant to cheating, test-wiseness, and socioeconomic test-prep variances ⁸ . Labeling unverified self-reports as "aptitude" is a fundamental psychometric error ⁸ .
Domain-Specific Self-Efficacy	Perceived capability to learn, persist through, and master fundamental academic challenges within specific streams (e.g., confidence in tackling complex mathematics) ⁸ .	Objective diagnostic mastery of Class 10 syllabus standards.	Social Cognitive Career Theory (SCCT) shows that interest development is mediated by self-efficacy beliefs and outcome expectations ⁶ . Measuring perceived efficacy provides vital context for whether expressed interest can translate into academic persistence ⁸ .
Personality Dispositions	Behavioral manifestations of work preferences (e.g., preference for structured	Clinical or trait-level personality diagnosis (e.g., full NEO-PI-R Big Five, MMPI clinical	Comprehensive personality assessment requires extensive item batteries

	procedural tasks vs. unstructured ideation) embedded directly within activity scenarios ⁹ .	scales, Myers-Briggs Type Indicator categorizations) ²⁴ .	(typically 60–240 items) ²⁴ . Personality profiling at age 15–16 is prone to developmental flux and carries the risk of harmful student labeling ²¹ .
Educational & Contextual Values	Unscored exploration of educational aspirations, parental career orientation, and subject preference hierarchies ⁶ .	Diagnosing socioeconomic class, assessing family dysfunction, or predicting commercial earning power.	Contextual parameters moderate whether a recommended stream is practically viable or socially supported, enabling personalized report generation without compromising psychometric scoring ⁷ .

Framework Comparison

Holland RIASEC Model

The RIASEC typology, formulated by John Holland, posits that vocational interests and occupational environments can be organized into six broad domains: Realistic (R), Investigative (I), Artistic (A), Social (S), Enterprising (E), and Conventional (C)⁵. Holland proposed that these domains exist along a hexagonal circumplex where adjacent types (e.g., Realistic–Investigative) share higher conceptual similarity than alternate or opposite types (e.g., Realistic–Social), establishing the principle of person–environment (P–E) congruence⁵.

The empirical literature provides extensive support for Holland's model, cementing its status as the most validated interest taxonomy in vocational psychology⁵. Meta-analyses synthesising decades of research indicate that congruence between an individual's interest profile and their chosen educational or work environment correlates significantly with job satisfaction (

$r = 0.21$ to 0.30), academic major stability, and long-term career persistence⁵.

Longitudinal studies evaluate interest stability across developmental transitions, demonstrating that while overall interest elevation fluctuates during adolescence, the rank-order stability of

RIASEC profiles reaches robust levels by age 15 to 17 ($r \approx 0.55$) and remains relatively stable across subsequent decades²⁰.

However, cross-cultural structural meta-analyses reveal that the circular order hypothesis (the

strict calculus assumption) frequently fails outside Western industrialized nations¹³. In developing economies, structural economic constraints, limited occupational exposure, and caste- or class-based prestige biases disrupt the symmetrical spacing between constructs⁷. In India specifically, Realistic occupations involving manual, trade, or physical labor carry historical social stigma, leading adolescents from diverse socioeconomic backgrounds to reject Realistic prompts regardless of their mechanical inclination⁷. Furthermore, Enterprising and Conventional dimensions often exhibit high multicollinearity in economies dominated by family-run retail, small-scale commerce, and administrative employment⁷. Consequently, while Holland's broad domains offer an indispensable starting taxonomy, direct importation of Western RIASEC instruments without structural adaptation is psychometrically unviable⁷.

Strong Interest Inventory

Originally constructed by E.K. Strong and periodically updated, the Strong Interest Inventory (SII) assesses vocational interests across General Occupational Themes (based on RIASEC), Basic Interest Scales (BIS), and Occupational Scales³¹.

The empirical validity of the SII is well established, particularly regarding its empirical criterion-keying methodology, which contrasts an individual's responses against the profiles of satisfied professionals within specific occupations²⁸. It has demonstrated high reliability and predictive validity for individuals aged 15 and older³².

Despite these strengths, the SII has severe operational limitations for Naaguru V1. Its Occupational Scales rely on proprietary, Western-normed datasets that do not map onto the Indian labor market or the four-track Intermediate system of Andhra Pradesh and Telangana¹. Furthermore, its administration length (approaching 300 items) imposes unsustainable cognitive fatigue on secondary school students using mobile devices⁹. The instrument also requires paid proprietary licensing, precluding its use as an open architectural baseline for a local ed-tech platform⁹.

ACT Interest Inventory and the O*NET Interest Profiler

Prediger adapted Holland's framework by identifying two bipolar cognitive axes underlying vocational interests: Data versus Ideas, and People versus Things²⁰. The United States Department of Labor applied these dimensional principles to construct the Occupational Information Network (O*NET) Interest Profiler (IP)⁹.

The 60-item O*NET Interest Profiler Short Form displays high psychometric rigor, exhibiting scale internal consistencies ranging from $\alpha = .80$ to .90, robust factorial stability across secondary and post-secondary student samples, and strong criterion validity in predicting occupational choices⁹. The inventory is public-domain, developed explicitly for self-directed educational and career exploration, and has been extensively validated on adolescent cohorts in secondary schools⁹.

The ONET Interest Profiler provides the most suitable structural starting point for Naaguru V1 because of its focus on concrete work and learning activities rather than abstract job titles⁹. Nevertheless, direct adoption without deep contextual modification introduces cultural bias¹⁹. Specific activities within the ONET inventory—such as operating industrial machinery, repairing

specialized appliances, or Western-style recreational activities—lack ecological validity for Class 10 students in South India, requiring comprehensive item localization⁹.

Indigenous Frameworks: The Cultural Preparation Process Model

Psychologist Gideon Arulmani introduced the Cultural Preparation Process Model and the Jiva career guidance methodology, developing a theoretical framework specifically grounded in Indian socio-cognitive realities⁷. Arulmani's empirical investigations across diverse Indian populations highlight fundamental departures from Western career developmental theory: First, career decision-making in India operates as a collective family livelihood enterprise rather than an individualistic pursuit of personal self-actualization⁷. Adolescents develop internalized "career beliefs" deeply influenced by familial survival needs, social prestige hierarchies, and community expectations⁷.

Second, Arulmani identifies a persistent "aptitude-interest asymmetry" in Indian adolescents⁸. External socialization frequently induces intense expressed interest in prestigious tracks (specifically engineering and medicine) despite low underlying affinity or foundational academic preparation, generating acute psychological distress and high attrition rates in higher education⁵.

Third, adolescent transitions in India are governed by institutional tracking gates¹. Students completing Class 10 do not select open occupational careers; they select rigid, state-mandated academic streams that dictate all subsequent professional options¹. Consequently, vocational assessment in India must evaluate task preferences that map onto academic disciplines rather than theoretical occupational archetypes⁸.

Synthesis and Framework Recommendation for Naaguru V1

Naaguru V1 adopts a hybrid conceptual architecture. From Holland and O*NET, the assessment incorporates the principle of measuring person-environment fit through concrete, task-level activity preferences evaluated along normalized interest dimensions⁵. From Arulmani's Cultural Preparation Process Model, the assessment integrates contextual career beliefs, family expectations, and localized academic streaming realities⁷.

The instrument organizes items around school-based learning tasks, practical problem-solving activities, and everyday community behaviors, avoiding social prestige bias and directly mapping interest profiles onto the four Intermediate streams of Andhra Pradesh and Telangana (MPC, BiPC, MEC, and CEC)¹.

Recommended Constructs/Dimensions

A critical review of the eight provisional dimensions initially proposed reveals significant conceptual redundancy, psychometric conflation, and vulnerability to social desirability bias⁹.

Evaluation of Provisional Dimensions

The dimension labeled "Analytical/problem solving" represents a cross-cutting cognitive competency rather than an isolated vocational interest domain. Problem-solving in symbolic mathematics is psychometrically distinct from problem-solving in interpersonal disputes, legal ethics, or ecological systems; retaining it as an independent dimension introduces construct contamination.

The provisional dimension "Scientific interest" fails to differentiate between physical sciences (which are heavily mathematical and mechanical) and biological sciences (which at the secondary level are largely descriptive, taxonomic, and physiological)³⁸.

"Numerical/quantitative interest" overlaps considerably with "Technology interest." Meanwhile, asking modern adolescents if they are interested in "technology" yields near-universal ceiling effects due to ubiquitous smartphone and consumer electronics usage, destroying psychometric discrimination. Technology interests must instead be operationalized through physical fabrication, mechanical systems, or formal algorithmic logic⁹.

The "Leadership/initiative" dimension is exceptionally prone to extreme social desirability bias among Indian youth, who face intense cultural pressure to display managerial ambition³⁷.

Furthermore, leadership is an organizational role and personality disposition, not a primary academic subject preference that differentiates MPC from CEC¹.

"Business/economic interest," "Social/people orientation," and "Creative interest" demonstrate strong construct validity, mapping cleanly onto established vocational taxonomies and regional academic streams¹.

The Operational Six-Dimension Model

To eliminate construct overlap while preserving full fidelity to the Intermediate academic streams of Andhra Pradesh and Telangana, Naaguru V1 establishes a streamlined six-dimension framework³.

Dimension Name	ID	Operational Psychometric Definition	Key Curricular & Behavioral Indicators	Direct Intermediate Stream Alignment
Investigative & Scientific Inquiry	ISI	Intrinsic curiosity regarding natural phenomena, biological organisms, chemical transformations, and empirical hypothesis testing ⁹ .	Observing living systems, isolating experimental variables, analyzing plant/animal physiology, investigating chemical reactions ⁹ .	Primary: BiPC [cite: 38] Secondary: MPC [cite: 3]
Quantitative &	QCR	Affinity for symbolic logic,	Formulating algebraic	Primary: MPC [cite: 3]

Computational Reasoning		numerical pattern recognition, algebraic deduction, algorithmic sequencing, and structured data manipulation ⁹ .	equations, deducing mathematical logic, tracing computational procedures, analyzing statistical charts ⁹ .	Secondary: MEC [cite: 3]
Technical & Mechanical Design	TMD	Preference for understanding physical mechanisms, electrical circuitry, spatial drafting, structural design, and hands-on fabrication ⁹ .	Deconstructing mechanical objects, drafting scale layouts, wiring electrical switches, building structural physical models ¹ .	Primary: MPC [cite: 1] Secondary: Polytechnic / Applied Tech [cite: 1]
Commercial & Economic Enterprise	CEE	Interest in financial markets, trade, pricing mechanisms, consumer dynamics, budgeting, and organizational resource optimization ¹ .	Calculating profit margins, analyzing supply and demand, pitching products, maintaining financial ledgers, tracking startups ¹ .	Primary: MEC / CEC [cite: 3] Secondary: Business & Commerce [cite: 41]
Social, Helping & Administrative Coordination	SHC	Orientation toward human development, community welfare, civic institutions,	Explaining concepts to peers, mediating conflicts, studying legal	Primary: CEC [cite: 3] Secondary: BiPC (Healthcare/A

		interpersonal mediation, public administration, and legal justice ³ .	rights and welfare policies, organizing civic service events ³ .	llied) [cite: 38]
Creative & Expressive Arts	CEA	Engagement with aesthetic composition, literary expression, storytelling, visual illustration, performing arts, and multimedia narrative analysis ⁹ .	Writing narrative stories and poetry, drawing visual layouts, critiquing cinematic staging, composing melodic rhythms ⁹ .	Primary: Humanities / Design / Media [cite: 1, 41] Secondary: Architecture / Literature [cite: 1, 41]

Interest vs. Aptitude vs. Personality

Rigorous psychometric practice requires maintaining unambiguous distinctions among psychological constructs⁸. Failing to separate affective interest from demonstrated aptitude or personality traits degrades measurement validity and can lead to harmful educational placement decisions⁵.

Construct	Operational Definition	Measurement Modality	Cognitive / Behavioral Target	Role in Naaguru V1 Guidance Engine
Aptitude	Latent cognitive capacity or potential to acquire proficiency in a specific skill domain when provided	Standardized, timed, objectively scored maximum-performance tests ⁸ .	Maximum cognitive processing speed, spatial reasoning, working memory capacity ⁸ .	Explicitly Excluded. Naaguru V1 does not administer timed maximum-performance tests ⁸ .

	structured training ⁸ .			
Ability	Current demonstrated mastery and execution of a knowledge base or practical skill ⁸ .	Standardized achievement exams, board test marks (SSC GPA) ⁸ .	Realized educational attainment and syllabus mastery ⁸ .	Contextual Benchmark. SSC performance used as self-reported contextual check, not test score ⁸ .
Interest	Trait-like affective preference to engage in specific activities, disciplines, and learning environments ⁹ .	Self-report typical-performance questionnaires using Likert scales ⁹ .	Intrinsic enjoyment, voluntary curiosity, sustained task engagement ⁹ .	Primary Scored Core. Forms the 36 scored items evaluating person-environment fit ⁹ .
Personality	Enduring, cross-situational patterns of thinking, feeling, behaving, and adapting ²³ .	Comprehensive trait inventories (e.g., Big Five, HEXACO) ²⁴ .	Behavioral style, emotional stability, conscientiousness, extraversion ²³ .	Secondary Task Framing. Embedded functionally within activity descriptions; no trait diagnosis ⁹ .
Values	Core cognitive representations of desirable end-states, goals, and guiding life principles ⁷ .	Forced-choice or normative work-value inventories ⁴⁴ .	Financial security, social prestige, personal autonomy, community service ⁷ .	Contextual Exploration. Addressed within unscored background modules ⁷ .
Work Preferences	Preferred physical	Environmental choice	Indoor desk-bound	Unscored Filter. Informs

	conditions, environmental contexts, and social modalities of work ⁵ .	scenarios ⁹ .	vs. outdoor mobile; collaborative vs. independent ⁵ .	personalized career area discovery ⁹ .
Self-Efficacy	Subjective confidence in one's personal capability to master domain-specific tasks and challenges ¹¹ .	Domain-specific confidence rating scales ¹¹ .	Perceived capability to pass difficult mathematics, learn biology, etc. ⁸ .	Integrated Component. Evaluated alongside task interest to assess persistence potential ⁸ .

Construct Strategy and Claims Boundaries for Naaguru V1

Naaguru V1 allocates its psychometric bandwidth exclusively to typical-performance measurement: approximately 90% of scored variance evaluates vocational interests (activity and learning enjoyment), while 10% incorporates domain-specific self-efficacy (perceived learning confidence)⁹. Cognitive aptitude testing is explicitly excluded⁸. Standardized aptitude measurement requires timed testing conditions, extensive normative calibration against national sample distributions, and security protocols against item exposure, none of which are viable within an exploratory, mobile-first experience⁸.

The guidance engine must maintain clear operational boundaries regarding its claims:

- The assessment must never label an endorsement of interest (e.g., "I enjoy solving numerical puzzles") as evidence of "High Mathematical Aptitude"⁸. Affective interest does not guarantee cognitive capability to master advanced calculus in Intermediate Mathematics Papers 1B and 2B³.
- The assessment must never claim to reveal a student's "inner genius," determine their "true career destiny," or predict board examination scores and future income⁵.
- The output report must explicitly communicate: *"This assessment reflects your current interests and learning preferences. Success in any intermediate stream also requires dedicated study habits, foundational practice, and academic perseverance"*⁵.

Adolescent Question Design Principles

Administering digital assessments to adolescents aged 15 to 17 requires specific psychometric adaptations to minimize cognitive load, satisficing, acquiescence bias, and mobile drop-off³⁷.

Reading Complexity and Readability Controls

- **Readability Ceiling:** Text must maintain a Flesch-Kincaid Grade Level (FKGL) between **4.0 and 5.5**, despite the target cohort completing Class 10¹⁷. Many students in regional English-medium schools display functional reading comprehension below their

chronological grade level, particularly when processing complex syntax².

- **Syntactic Structure:** Stems must not exceed 12 to 16 words. Subordinate clauses, passive voice constructions, and conditional clauses must be systematically removed.
- **Lexical Accessibility:** Technical academic terminology (e.g., "empirical methodology," "macroeconomic indicators," "aesthetic composition") must be converted into concrete behavioral equivalents (e.g., "doing experiments," "how prices change," "drawing designs")¹⁸.

Item Format: Rating Scales versus Forced Choice

Psychometric Evaluation Criterion	5-Point Likert Enjoyment Scale	Forced-Choice / Ipsative Format	Tactical Recommendation for Naaguru V1
Cognitive Load on Mobile	Low to moderate. Evaluates a single, isolated prompt against a clear continuum ¹⁹ .	High. Demands simultaneous cognitive comparison of disparate tasks across multiple dimensions ²⁴ .	Use the 5-point Likert enjoyment scale ¹⁹ . Forced-choice items induce decision fatigue on mobile screens, increasing incomplete submissions among adolescents ²⁴ .
Scale Properties & Measurement	Normative. Generates independent construct scores, permitting normative comparison across peers and absolute profile elevation ²⁴ .	Ipsative. Generates zero-sum relative rankings; individual scales cannot be compared normatively or across individuals ²⁴ .	Normative scoring is critical for educational streaming. An ipsative format cannot determine whether a student has high affinity across all fields or uniformly low interest across all fields ²⁴ .
Vulnerability to Response Bias	Susceptible to acquiescence bias (yea-saying) and	Controls for acquiescence bias, but introduces	Deploy the 5-point Likert scale with concrete

	central tendency bias ³⁷ .	artificial trade-offs that frustrate test-takers ²⁴ .	behavioral anchors and varied activity contexts to minimize acquiescence ¹⁹ .
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- **Standardized Anchor Continuum:**
 - 1 = Strongly Dislike / Not Interested At All
 - 2 = Dislike / Slightly Uninterested
 - 3 = Neutral / Unsure
 - 4 = Like / Interested
 - 5 = Strongly Like / Extremely Interested

Mitigating Measurement Errors and Artifacts

- **Double-Barrelled Items:** Each item stem must isolate a single cognitive or behavioral activity. Prompts that combine disparate activities (e.g., "Do you enjoy repairing bicycles and writing software?") distort construct measurement and must be split⁹.
- **Leading Stems and Prestige Biases:** Stems must avoid incorporating prestigious professional titles or socially elevated descriptors. Stems referencing "executive leadership" or "expert surgeon" elicit prestige-seeking responses; prompts must evaluate the underlying daily task (e.g., "organizing a team project" or "learning human anatomy")⁸.
- **Socioeconomic Neutrality:** Scenarios must never assume affluent domestic resources, personal computer ownership, or expensive hobbies. Activities requiring high-end laptops, international travel, or robotics kits must be replaced with scenarios centered on universal school activities, household items, or everyday community observations¹⁸.

Pacing, Engagement, and Fatigue Mitigation

Field data indicates that adolescent attention on mobile assessments declines sharply after 8 to 10 minutes⁹.

- **Item Count and Administration Time:** An inventory composed of **36 scored items and 4 unscored contextual items (40 items total)** requires approximately 6 to 8 minutes, with students spending an average of 10 to 12 seconds per item.
- **Mobile Experience Design:** Items should be presented one per screen with large tap targets and automatic advancement upon response selection⁹. A persistent progress bar and brief encouraging transition cards placed every 10 items sustain engagement without inflating scores⁵⁴.

Indian / AP / Telangana Considerations

Educational decision-making among secondary school students in Andhra Pradesh and Telangana is shaped by a distinct institutional and cultural landscape². Research in Indian educational sociology and vocational psychology indicates several key dynamics that must

inform assessment design¹.

The Engineering-Medicine Coaching Duopoly

The secondary and higher secondary educational landscape in Andhra Pradesh and Telangana is shaped by a powerful duopoly centered on engineering and medicine¹. Large corporate junior college networks recruit heavily across Class 10 cohorts, promoting residential and integrated coaching programs focused on national entrance examinations (IIT-JEE and NEET) and state entrance tests (AP/TS EAPCET)¹.

As a result, many students internalize high perceived interest in science tracks due to parental pressure, institutional promotion, and peer conformity, rather than genuine intrinsic interest⁵. The assessment must neutralize this bias by decoupling foundational scientific and quantitative inquiry from high-prestige titles, evaluating raw task engagement rather than the social allure of becoming a "software engineer" or "doctor"⁵.

Collectivism and Livelihood Realities

In accordance with Arulmani's Cultural Preparation Process Model, vocational choices in India reflect collective family livelihood strategies rather than individualistic personal exploration⁷.

- **First-Generation Learners and Rural-Urban Divides:** In rural districts and semi-urban mandals, first-generation college aspirants frequently lack familiarity with emerging digital economy careers (e.g., product management, data visualization, UI/UX design, public policy analysis)². Guidance items must be anchored in universal physical, social, and academic experiences accessible across diverse demographic groups¹⁸.
- **Destigmatizing Commerce and Humanities:** State board enrollment figures show that MEC and CEC tracks are often perceived as fallback streams for students unable to handle the academic rigor of science¹. Naaguru's assessment architecture must counter this perception by presenting commerce, economics, and civics as intellectually challenging and rewarding pathways that lead to careers in corporate law, chartered accountancy, public administration, and entrepreneurship¹.

English + Telugu Assessment Design

Providing an assessment in both English and Telugu requires following the International Test Commission (ITC) Guidelines for Translating and Adapting Tests (Second Edition, 2017) to guarantee construct equivalence¹⁰. Literal, word-for-word translation frequently alters construct meaning, shifts item difficulty, or introduces cultural ambiguity¹⁰.

Linguistic and Structural Considerations

- **Syntactic Realignment:** English follows a Subject-Verb-Object (SVO) order, whereas Telugu uses a Subject-Object-Verb (SOV) structure with complex agglutinative verb conjugations. Direct mechanical translations often result in awkward phrasing; items must be rewritten to match natural Telugu syntax while preserving the underlying psychological construct¹⁰.
- **Register Selection (*Vyavaharika* vs. *Grandhika*):** School textbooks sometimes feature archaic literary Telugu (*Grandhika Bhasha*), which creates comprehension barriers for adolescents in spoken contexts. The assessment must use contemporary standard

educational Telugu (*Vyavaharika Bhasha*), matching the natural register used in secondary school classrooms¹⁰.

- **Handling Technical Terminology and Loanwords:** Adolescents in Telugu-medium schools regularly use common English loanwords for technical and everyday concepts⁵³. Forcing obscure classical Telugu translations harms comprehension. Terms such as "Computer" (కంప్యూటర్) are preferred over "Sanganaka Yantramu" (సంగణక యంత్రము), and "Experiment" (ప్రయోగం) or "Project" (ప్రాజెక్ట్) should be rendered using standard conversational loanwords⁵³.

Operational Adaptation Protocol

The adaptation process follows a structured six-stage workflow adhering to ITC standards¹⁰:

1. **Dual Independent Forward Translation:** Two bilingual translators independently translate items from English into Telugu. One translator is an educational psychologist familiar with vocational interest constructs; the second is a native Telugu speaker unfamiliar with psychometric testing, ensuring natural linguistic flow¹⁰.
2. **Expert Panel Reconciliation:** A panel comprising a Telugu linguist, a psychometrician, and an experienced secondary school educator reviews both translations, resolving discrepancies to establish Consensus Version A¹⁰.
3. **Blind Back-Translation:** Two independent bilingual translators, blinded to the original English items and psychometric objectives, translate Consensus Version A back into English¹⁰.
4. **Discrepancy Analysis and Equivalence Review:** The panel compares the back-translated English text with the original source text to identify semantic drift, shifts in item difficulty, or lost subtleties¹⁰.
5. **Cognitive Debriefing with Adolescents:** Conduct cognitive "think-aloud" interviews with 15 bilingual Class 10 students across Hyderabad, Guntur, and Mahbubnagar to verify that both language versions convey identical meaning and emotional tone¹⁶.
6. **Empirical Measurement Invariance and DIF Testing:** Following pilot testing, conduct Differential Item Functioning (DIF) analyses across language groups to confirm that items function equivalently across English and Telugu administrations¹⁰.

Stream Mapping (MPC, BiPC, MEC, CEC)

The four primary Intermediate academic streams governed by BIEAP and TSBIE require specific combinations of four core subjects in addition to English and a second language (such as Telugu, Hindi, Sanskrit, or Urdu)³. Mapping psychometric interest dimensions to these streams requires distinguishing empirical evidence from operational product assumptions¹.

Intermediate Stream	Core Board Subject Curriculum	Typical Career Pathways	(A) Empirically Supported Relationships	(B) Operational Product Assumptions	(C) Attributes That Cannot Be

			ps	ns	Inferred
MPC	• Mathematics 1A & 1B (Year 1) • Mathematics 2A & 2B (Year 2) • Physics • Chemistry³	Software engineering, mechanical/electrical engineering, data science, defense (NDA), architecture, physical sciences ¹ .	High intrinsic interest in abstract quantitative reasoning (QCR) and technical-mechanical systems (TMD) strongly correlates with persistence in engineering and physical science tracks ⁹ .	Students scoring high in both QCR and TMD will find the MPC curriculum intellectually stimulating and aligned with their preferred learning tasks ³ .	• Ability to clear highly competitive JEE Advanced or EAPCET cutoffs. • Tolerance for 14-hour daily rote coaching routines. • Innate programming talent⁸.
BiPC	• Botany • Zoology • Physics • Chemistry³	Medicine (MBBS/BDS), pharmacy, veterinary science, agricultural research, biotechnology, allied health sciences (nursing, radiology) ³⁸ .	Strong interest in natural phenomena and biological systems (ISI), combined with social-helping orientation (SHC), predicts persistence in life sciences	Students with elevated ISI and moderate-to-high SHC are well-suited to the intense anatomical and taxonomic study required in Botany and Zoology ³⁸ .	• Emotional resilience required for clinical medical practice. • Probability of securing a government medical seat via NEET. • Innate surgical dexterity⁸.

			and clinical environments ⁹ .		
MEC	- Mathematics 1A & 1B (Year 1) - Mathematics 2A & 2B (Year 2) - Commerce - Economics³	Chartered Accountancy (CA), financial analytics, corporate law, actuary, business management (BBA), economic research ¹ .	Co-elevated quantitative interest (QCR) and commercial enterprise interest (CEE) predict satisfaction in finance, actuarial science, and economic planning ⁹ .	Students who enjoy mathematics but prefer commercial and organizational applications over physical chemistry and physics thrive in MEC ¹ .	- Ability to pass rigorous multi-stage CA foundation and intermediate examinations. - Innate entrepreneurial risk tolerance. - Executive management aptitude⁸.
CEC	- Commerce - Economics - Civics (Political Science)³	Civil Services (UPSC/State PSC), law (CLAT), public administration, corporate banking, journalism, human resources ¹ .	Strong interest in civic systems and social coordination (SHC) paired with commercial enterprise (CEE) correlates with persistence in law, public	Students who demonstrate lower affinity for advanced mathematics but strong enthusiasm for societal affairs, law, trade, and governance are best served by CEC ¹ .	- Competitive success in UPSC/State Civil Service exams. - Legal argumentation and courtroom litigation capability. - Academic stamina in

			administrati on, and social sciences ⁹ .		descriptive, reading-he avy disciplines ⁸ .
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Responsible Career Recommendation Design

Communicating psychometric findings to adolescents and their parents requires strict avoidance of deterministic language⁵. Phrases such as "*You are destined to become an engineer*" or "*You are unsuited for the science stream*" are psychometrically unfounded and educationally harmful⁵.

The Need for Exploratory Framing

Vocational interests during adolescence are still forming; longitudinal research indicates that while rank-order stability is notable, interest profiles continue to evolve through post-secondary schooling²⁰. Furthermore, an interest assessment evaluates affective attraction to tasks, not cognitive ability or future academic achievement⁸. Reporting must therefore present findings as probabilistic pathways for self-reflection and guided exploration⁵.

Inappropriate / Deterministic Language	Responsible / Exploratory Communication Pattern	Psychometric Justification
<i>"You must take MPC because your analytical score is in the 95th percentile."</i>	<i>"Your responses show a strong interest in solving logic puzzles, working with numbers, and understanding how machines operate. Because of these interests, the MPC stream is an excellent option for you to explore."</i>	Frames recommendations around observable student interests rather than definitive prescriptions, leaving agency with the student and family ⁵ .
<i>"You scored low in science. You are completely unfit for BiPC or medicine."</i>	<i>"Your responses suggest you may find less enjoyment in memorizing biological systems or doing laboratory experiments compared to other areas. If you are considering BiPC, talk with your science teacher about the daily</i>	Prevents fatalistic foreclosure and encourages constructive dialogue with educators, acknowledging that low interest can be re-evaluated ⁸ .

	study requirements."	
"Your assessment results prove that you will become a successful Chartered Accountant."	"Careers in finance, accounting, and economic planning—such as Chartered Accountancy—align closely with the activities you find interesting. Consider exploring what a CA or business analyst does on a typical workday."	Emphasizes person-environment congruence over predictive certainty, connecting interests to daily occupational tasks ⁵ .

Report Structure

1. **Primary Interest Dimensions:** A visual profile highlighting the individual's top two or three interest domains, described in accessible, strengths-oriented language⁵.
2. **Intermediate Stream Alignment Exploration:** Displays alignment across MPC, BiPC, MEC, and CEC streams using non-ranked, qualitative fit categories (e.g., "Strong Affinity," "Moderate Affinity," "Worth Exploring"), deliberately avoiding misleading numerical percentages¹².
3. **Exploratory Action Roadmap:** Specific next steps for the student, such as reviewing Class 11 textbooks (e.g., BIEAP Maths 1A or Botany), speaking with professionals in related fields, and discussing results with family to align interests with educational plans⁷.

Validity & Reliability Requirements

Building a psychometrically defensible guidance instrument requires adhering to established psychometric standards for score reliability and construct validity⁹.

Psychometric Definitions and Measurement Standards

- **Reliability:** The overall consistency and precision of measurement scores across repeated administrations and within test components⁹.
 - *Internal Consistency:* The degree to which test items within a single construct scale evaluate the same underlying psychological dimension. Evaluated via Cronbach's alpha (α) and McDonald's omega (ω). Target standard: $\alpha \geq$ for exploratory educational guidance; $\alpha \geq 0.85$ for high-stakes screening⁹.
 - *Test-Retest Reliability:* The temporal stability of assessment scores over a defined time interval (e.g., 14 days). Target standard: Pearson $r \geq$ among secondary school adolescents⁹.
- **Validity:** The degree to which empirical evidence and theoretical rationale support the adequacy and appropriateness of interpretations based on test scores³³.

- *Construct Validity*: Verification that the instrument measures the hypothesized theoretical dimensions (e.g., QCR, ISI) rather than confounding variables (such as reading speed or social desirability). Assessed using Confirmatory Factor Analysis (CFA)⁹.
- *Content Validity*: Systematic confirmation by subject-matter experts that the item pool comprehensively and accurately samples the target behavioral domains¹⁶.
- *Criterion Validity*: Evidence demonstrating that scores correlate systematically with external educational benchmarks, including concurrent validity (correlation with self-reported subject preferences) and predictive validity (stream persistence and satisfaction 12 months later)⁹.

Responsible Boundaries for Naaguru V1

- **Permissible Claims:**

- *"The Naaguru assessment provides a reliable profile of your self-reported vocational interests and learning preferences across six broad educational domains"*⁹.
- *"Our recommendations reflect person-environment congruence, identifying Intermediate streams and career fields where your daily learning tasks align with your preferred activities"*¹⁵.

- **Impermissible Claims:**

- The platform must NOT claim to evaluate general intelligence (g), cognitive ability, or innate academic talent⁸.
- The platform must NOT claim to diagnose personality disorders, emotional stability, or clinical conditions²⁴.
- The platform must NOT guarantee entrance examination rank outcomes (e.g., *"Guaranteed to clear NEET/JEE if you enroll in this stream"*)⁸.
- The platform must NOT assert predictive certainty regarding future earnings or career success⁸.

Practical Pilot and Validation Roadmap

The validation process progresses through eight systematic phases:

- **Phase 1 (Qualitative Review)**: Content validity review by a panel of 5 experts (2 educational psychologists, 2 senior secondary teachers, 1 Telugu linguist) to confirm developmental appropriateness and translation fidelity¹⁶.
- **Phase 2 (Cognitive Debriefing)**: Think-aloud interviews with 20 Class 10 students across Hyderabad, Guntur, and Mahbubnagar to evaluate reading ease, vocabulary comprehension, and item interpretation¹⁶.
- **Phase 3 (Alpha Pilot)**: Administration of the full 64-item candidate pool to $N =$ students across four partner schools to analyze response variance and initial scale cohesion⁹.
- **Phase 4 (Item Refinement)**: Psychometric analysis of alpha pilot data, removing or revising items showing item-total correlations $r_{it} < 0.30$ or skewness $> |1.5|$ to

select the final 40-item operational battery⁹.

- **Phase 5 (Beta Field Pilot):** Large-scale field testing with $N =$ Class 10 students to establish regional norms, evaluate internal consistency ($\alpha \geq$), and confirm the six-factor model via CFA ($CFI \geq$, $RMSEA \leq$)⁹.
- **Phase 6 (Cross-Language DIF Analysis):** Differential Item Functioning analysis across bilingual sub-samples ($N =$ English, $N =$ Telugu) to ensure measurement equivalence¹⁰.
- **Phase 7 (Test-Retest Stability):** Re-administering the assessment to $N =$ students after a 14-day interval to verify temporal stability ($r \geq$)⁹.
- **Phase 8 (Longitudinal Tracking):** Following $N =$ students into their first year of Intermediate education to evaluate predictive validity and satisfaction with their chosen stream⁹.

Recommended Assessment Architecture

The operational assessment architecture balances psychometric coverage with the practical constraints of mobile administration, resulting in a 40-item total battery⁹.

Structural Specifications

- **Psychometric Scored Core (36 Items):** Exactly 6 items per construct across the six core dimensions (ISI, QCR, TMD, CEE, SHC, CEA). Items are distributed across four thematic sections to vary task contexts and prevent response fatigue⁹.
- **Response Format:** A 5-point Likert enjoyment scale ranging from 1 ("Strongly Dislike / Not Interested At All") to 5 ("Strongly Like / Extremely Interested")¹⁹.
- **Unscored Contextual Module (4 Items):** Four single-select categorical items positioned at onboarding and exit. These items capture baseline academic ease, family career expectations, educational aspiration horizons, and workplace environment preferences, providing contextual filters for report generation without distorting construct scores⁶.

Operational Section	Item Positions	Primary Thematic Focus	Included Item Identifiers
Onboarding Context	Items 1–2	Baseline Academic Context & Family Expectations	CTX-01 (Subject Ease), CTX-02 (Family Expectation)
Section 1: Learning & Class Tasks	Items 3–12	Academic Activities, Experiments &	ISI-01, QCR-02, TMD-03, CEE-01, SHC-01, CEA-01,

		Calculations	ISI-02, QCR-01, TMD-02, CEE-02
Section 2: Practical Problem-Solving	Items 13–21	Hands-on Mechanical, Design & Numerical Tasks	SHC-02, CEA-02, ISI-04, QCR-03, TMD-01, CEE-04, SHC-03, CEA-03, QCR-04
Section 3: Real-World Systems & Markets	Items 22–30	Practical Science, Trade, Civics & Community	TMD-04, CEE-03, SHC-04, CEA-05, ISI-03, QCR-05, TMD-05, CEE-05, SHC-05
Section 4: Creative & Expressive Projects	Items 31–38	Construction, Community Welfare & Arts	CEA-04, ISI-06, QCR-06, TMD-06, CEE-06, SHC-06, CEA-06, ISI-08
Exit Context	Items 39–40	Future Horizons & Workplace Preferences	CTX-03 (Degree Horizon), CTX-04 (Work Setting)

Candidate Question Bank

The 64-item candidate pool includes 60 scored interest items (10 per construct across the six core dimensions) and 4 unscored contextual items, providing a thoroughly vetted pool for pilot testing and operational selection⁹.

Investigative & Scientific Inquiry (ISI)

Item ISI-01

- **Construct Measured:** Investigative & Scientific Inquiry (ISI)
- **Question:** How much do you enjoy using a microscope or magnifying lens to examine plant leaves, insects, or tiny pond water organisms?
- **Answer Choices:** (1) Strongly Dislike (2) Dislike (3) Neutral / Unsure (4) Like (5) Strongly Like
- **Question Type:** 5-point Likert Enjoyment Scale
- **Why It Measures the Construct:** Taps curiosity regarding biological structures and microscopic observation, central to Botany and Zoology⁹.

- **Scoring Direction:** Positive ($1 \rightarrow 5$)
- **Potential Ambiguity:** Minimal; references familiar school equipment.
- **Cultural/Context Concerns:** Basic lenses are universal across urban and rural schools¹⁸.
- **Reading Difficulty Estimate:** FKGL 4.8
- **Class 10 Suitability:** Fully suitable.
- **Pilot Requirement:** Include in alpha pilot.

Item ISI-02

- **Construct Measured:** Investigative & Scientific Inquiry (ISI)
- **Question:** How much do you enjoy mixing household liquids (like lemon juice, baking soda, and vinegar) to observe chemical reactions and color changes?
- **Answer Choices:** (1) Strongly Dislike (2) Dislike (3) Neutral / Unsure (4) Like (5) Strongly Like
- **Question Type:** 5-point Likert Enjoyment Scale
- **Why It Measures the Construct:** Captures curiosity regarding chemical reactions and molecular changes⁹.
- **Scoring Direction:** Positive ($1 \rightarrow 5$)
- **Potential Ambiguity:** None.
- **Cultural/Context Concerns:** Uses everyday household ingredients, avoiding assumptions of advanced laboratory access³⁹.
- **Reading Difficulty Estimate:** FKGL 4.2
- **Class 10 Suitability:** Fully suitable.
- **Pilot Requirement:** Include in alpha pilot.

Item ISI-03

- **Construct Measured:** Investigative & Scientific Inquiry (ISI)
- **Question:** How much do you enjoy reading articles or watching videos about how human vaccines, medicines, and the immune system fight infections?
- **Answer Choices:** (1) Strongly Dislike (2) Dislike (3) Neutral / Unsure (4) Like (5) Strongly Like
- **Question Type:** 5-point Likert Enjoyment Scale
- **Why It Measures the Construct:** Measures interest in medical biology, human immunology, and health sciences⁹.
- **Scoring Direction:** Positive ($1 \rightarrow 5$)
- **Potential Ambiguity:** Minimal.
- **Cultural/Context Concerns:** Highly relatable following public health campaigns³⁸.
- **Reading Difficulty Estimate:** FKGL 5.2
- **Class 10 Suitability:** Fully suitable.
- **Pilot Requirement:** Include in alpha pilot.

Item ISI-04

- **Construct Measured:** Investigative & Scientific Inquiry (ISI)
- **Question:** How much do you enjoy tracking why a plant in your garden is drying up by

testing different soil types and watering schedules?

- **Answer Choices:** (1) Strongly Dislike (2) Dislike (3) Neutral / Unsure (4) Like (5) Strongly Like
- **Question Type:** 5-point Likert Enjoyment Scale
- **Why It Measures the Construct:** Evaluates systematic scientific method, variable isolation, and botanical inquiry⁹.
- **Scoring Direction:** Positive (**1 → 5**)
- **Potential Ambiguity:** None.
- **Cultural/Context Concerns:** Highly relatable across semi-urban and agrarian rural communities².
- **Reading Difficulty Estimate:** FKGL 4.5
- **Class 10 Suitability:** Fully suitable.
- **Pilot Requirement:** Include in alpha pilot.

Item ISI-05

- **Construct Measured:** Investigative & Scientific Inquiry (ISI)
- **Question:** How much do you enjoy learning about planets, stars, black holes, and the laws of gravity governing outer space?
- **Answer Choices:** (1) Strongly Dislike (2) Dislike (3) Neutral / Unsure (4) Like (5) Strongly Like
- **Question Type:** 5-point Likert Enjoyment Scale
- **Why It Measures the Construct:** Assesses theoretical astrophysics and physical science inquiry, aligned with Physics³.
- **Scoring Direction:** Positive (**1 → 5**)
- **Potential Ambiguity:** May capture passive entertainment interest rather than active scientific curiosity.
- **Cultural/Context Concerns:** Highly visible topic due to ISRO media coverage; monitor for ceiling effects¹.
- **Reading Difficulty Estimate:** FKGL 5.0
- **Class 10 Suitability:** Fully suitable.
- **Pilot Requirement:** Test in pilot for discrimination parameters.

Item ISI-06

- **Construct Measured:** Investigative & Scientific Inquiry (ISI)
- **Question:** How much do you enjoy dissecting flower petals and seeds to understand their inner reproductive parts?
- **Answer Choices:** (1) Strongly Dislike (2) Dislike (3) Neutral / Unsure (4) Like (5) Strongly Like
- **Question Type:** 5-point Likert Enjoyment Scale
- **Why It Measures the Construct:** Directly measures botanical morphology and hands-on laboratory observation³⁸.
- **Scoring Direction:** Positive (**1 → 5**)

- **Potential Ambiguity:** None.
- **Cultural/Context Concerns:** Focuses on plant structures, avoiding cultural hesitations around animal dissection³⁸.
- **Reading Difficulty Estimate:** FKGL 4.9
- **Class 10 Suitability:** Fully suitable.
- **Pilot Requirement:** Include in alpha pilot.

Item ISI-07

- **Construct Measured:** Investigative & Scientific Inquiry (ISI)
- **Question:** How much do you enjoy testing how water flows through different types of sand, clay, and rock layers?
- **Answer Choices:** (1) Strongly Dislike (2) Dislike (3) Neutral / Unsure (4) Like (5) Strongly Like
- **Question Type:** 5-point Likert Enjoyment Scale
- **Why It Measures the Construct:** Measures geological and environmental science inquiry⁹.
- **Scoring Direction:** Positive ($1 \rightarrow 5$)
- **Potential Ambiguity:** None.
- **Cultural/Context Concerns:** Directly aligned with Class 10 Earth science topics¹⁸.
- **Reading Difficulty Estimate:** FKGL 4.3
- **Class 10 Suitability:** Fully suitable.
- **Pilot Requirement:** Include in candidate reserve pool.

Item ISI-08

- **Construct Measured:** Investigative & Scientific Inquiry (ISI)
- **Question:** How much do you enjoy learning the biological reasons behind why animals behave differently during seasonal migrations or weather changes?
- **Answer Choices:** (1) Strongly Dislike (2) Dislike (3) Neutral / Unsure (4) Like (5) Strongly Like
- **Question Type:** 5-point Likert Enjoyment Scale
- **Why It Measures the Construct:** Measures interest in zoology, animal behavior, and ecological adaptation⁹.
- **Scoring Direction:** Positive ($1 \rightarrow 5$)
- **Potential Ambiguity:** Minimal.
- **Cultural/Context Concerns:** Universally accessible natural science concept¹⁸.
- **Reading Difficulty Estimate:** FKGL 5.3
- **Class 10 Suitability:** Fully suitable.
- **Pilot Requirement:** Include in alpha pilot.

Item ISI-09

- **Construct Measured:** Investigative & Scientific Inquiry (ISI)
- **Question:** How much do you enjoy investigating the chemical pH levels of drinking water using indicator paper to see if it is safe?
- **Answer Choices:** (1) Strongly Dislike (2) Dislike (3) Neutral / Unsure (4) Like (5) Strongly

Like

- **Question Type:** 5-point Likert Enjoyment Scale
- **Why It Measures the Construct:** Assesses experimental chemistry inquiry and practical scientific measurement⁹.
- **Scoring Direction:** Positive ($1 \rightarrow 5$)
- **Potential Ambiguity:** Indicator paper must be clearly explained in Telugu translation.
- **Cultural/Context Concerns:** Part of the Class 10 syllabus (Acids, Bases and Salts)³⁹.
- **Reading Difficulty Estimate:** FKGL 5.1
- **Class 10 Suitability:** Fully suitable.
- **Pilot Requirement:** Include in candidate reserve pool.

Item ISI-10

- **Construct Measured:** Investigative & Scientific Inquiry (ISI)
- **Question:** How much do you enjoy finding out why certain genetic traits (like eye color or curly hair) pass from grandparents to children?
- **Answer Choices:** (1) Strongly Dislike (2) Dislike (3) Neutral / Unsure (4) Like (5) Strongly Like
- **Question Type:** 5-point Likert Enjoyment Scale
- **Why It Measures the Construct:** Assesses curiosity regarding genetics and biological inheritance¹⁸.
- **Scoring Direction:** Positive ($1 \rightarrow 5$)
- **Potential Ambiguity:** None.
- **Cultural/Context Concerns:** Highly relatable and personally relevant across backgrounds¹⁸.
- **Reading Difficulty Estimate:** FKGL 4.7
- **Class 10 Suitability:** Fully suitable.
- **Pilot Requirement:** Include in candidate reserve pool.

Quantitative & Computational Reasoning (QCR)

Item QCR-01

- **Construct Measured:** Quantitative & Computational Reasoning (QCR)
- **Question:** How much do you enjoy finding patterns in sequences of numbers or solving sudoku and logic number grids?
- **Answer Choices:** (1) Strongly Dislike (2) Dislike (3) Neutral / Unsure (4) Like (5) Strongly Like
- **Question Type:** 5-point Likert Enjoyment Scale
- **Why It Measures the Construct:** Evaluates intrinsic enjoyment of numerical patterns and inductive mathematical reasoning⁹.
- **Scoring Direction:** Positive ($1 \rightarrow 5$)
- **Potential Ambiguity:** Pairing "sudoku" with "number patterns" ensures familiarity in rural settings³⁹.
- **Cultural/Context Concerns:** Connects directly with Class 10 Progressions¹⁸.

- **Reading Difficulty Estimate:** FKGL 4.6
- **Class 10 Suitability:** Fully suitable.
- **Pilot Requirement:** Include in alpha pilot.

Item QCR-02

- **Construct Measured:** Quantitative & Computational Reasoning (QCR)
- **Question:** When faced with a word problem in math, how much do you enjoy translating it into algebraic equations to find the unknown value?
- **Answer Choices:** (1) Strongly Dislike (2) Dislike (3) Neutral / Unsure (4) Like (5) Strongly Like
- **Question Type:** 5-point Likert Enjoyment Scale
- **Why It Measures the Construct:** Measures affinity for formal mathematical modeling and symbolic translation³.
- **Scoring Direction:** Positive ($1 \rightarrow 5$)
- **Potential Ambiguity:** None; standard secondary mathematics task³⁹.
- **Cultural/Context Concerns:** Strong indicator of readiness for MPC/MEC curricula¹.
- **Reading Difficulty Estimate:** FKGL 5.0
- **Class 10 Suitability:** Fully suitable.
- **Pilot Requirement:** Include in alpha pilot.

Item QCR-03

- **Construct Measured:** Quantitative & Computational Reasoning (QCR)
- **Question:** How much do you enjoy writing down step-by-step instructions (algorithms) so someone else can solve a complex riddle without making mistakes?
- **Answer Choices:** (1) Strongly Dislike (2) Dislike (3) Neutral / Unsure (4) Like (5) Strongly Like
- **Question Type:** 5-point Likert Enjoyment Scale
- **Why It Measures the Construct:** Evaluates algorithmic thinking and procedural logic without requiring prior coding experience⁹.
- **Scoring Direction:** Positive ($1 \rightarrow 5$)
- **Potential Ambiguity:** Provide a clear Telugu gloss for "algorithm."
- **Cultural/Context Concerns:** Device-independent; equitable across urban and rural cohorts⁵³.
- **Reading Difficulty Estimate:** FKGL 5.2
- **Class 10 Suitability:** Fully suitable.
- **Pilot Requirement:** Include in alpha pilot.

Item QCR-04

- **Construct Measured:** Quantitative & Computational Reasoning (QCR)
- **Question:** How much do you enjoy calculating speed, time, and distance to figure out the fastest travel route between two towns?
- **Answer Choices:** (1) Strongly Dislike (2) Dislike (3) Neutral / Unsure (4) Like (5) Strongly Like
- **Question Type:** 5-point Likert Enjoyment Scale

- **Why It Measures the Construct:** Measures practical applied quantitative problem-solving and kinematic modeling⁹.
- **Scoring Direction:** Positive ($1 \rightarrow 5$)
- **Potential Ambiguity:** None.
- **Cultural/Context Concerns:** Familiar everyday problem across regional geography².
- **Reading Difficulty Estimate:** FKGL 4.5
- **Class 10 Suitability:** Fully suitable.
- **Pilot Requirement:** Include in alpha pilot.

Item QCR-05

- **Construct Measured:** Quantitative & Computational Reasoning (QCR)
- **Question:** How much do you enjoy organizing scattered facts or exam marks into neat data tables, bar charts, and pie graphs?
- **Answer Choices:** (1) Strongly Dislike (2) Dislike (3) Neutral / Unsure (4) Like (5) Strongly Like
- **Question Type:** 5-point Likert Enjoyment Scale
- **Why It Measures the Construct:** Assesses data structuring, descriptive statistics interest, and graphical representation³.
- **Scoring Direction:** Positive ($1 \rightarrow 5$)
- **Potential Ambiguity:** None.
- **Cultural/Context Concerns:** Directly aligned with Class 10 Statistics and Economics¹⁸.
- **Reading Difficulty Estimate:** FKGL 4.8
- **Class 10 Suitability:** Fully suitable.
- **Pilot Requirement:** Include in alpha pilot.

Item QCR-06

- **Construct Measured:** Quantitative & Computational Reasoning (QCR)
- **Question:** How much do you enjoy calculating probability, such as figuring out the exact chances of winning a game with dice or coins?
- **Answer Choices:** (1) Strongly Dislike (2) Dislike (3) Neutral / Unsure (4) Like (5) Strongly Like
- **Question Type:** 5-point Likert Enjoyment Scale
- **Why It Measures the Construct:** Evaluates affinity for probability theory, risk calculation, and combinatorics³.
- **Scoring Direction:** Positive ($1 \rightarrow 5$)
- **Potential Ambiguity:** None.
- **Cultural/Context Concerns:** Direct curricular connection to Class 10 Probability⁴⁰.
- **Reading Difficulty Estimate:** FKGL 4.7
- **Class 10 Suitability:** Fully suitable.
- **Pilot Requirement:** Include in alpha pilot.

Item QCR-07

- **Construct Measured:** Quantitative & Computational Reasoning (QCR)

- **Question:** How much do you enjoy using geometric formulas to calculate the exact volume and surface area of unusual containers or buildings?
- **Answer Choices:** (1) Strongly Dislike (2) Dislike (3) Neutral / Unsure (4) Like (5) Strongly Like
- **Question Type:** 5-point Likert Enjoyment Scale
- **Why It Measures the Construct:** Assesses geometric visualization, mensuration, and spatial calculation⁹.
- **Scoring Direction:** Positive ($1 \rightarrow 5$)
- **Potential Ambiguity:** None.
- **Cultural/Context Concerns:** Standard topic in Class 10 Mensuration¹⁸.
- **Reading Difficulty Estimate:** FKGL 5.1
- **Class 10 Suitability:** Fully suitable.
- **Pilot Requirement:** Include in candidate reserve pool.

Item QCR-08

- **Construct Measured:** Quantitative & Computational Reasoning (QCR)
- **Question:** When a computer game or phone app behaves unexpectedly, how much do you enjoy figuring out the exact rule or glitch causing the error?
- **Answer Choices:** (1) Strongly Dislike (2) Dislike (3) Neutral / Unsure (4) Like (5) Strongly Like
- **Question Type:** 5-point Likert Enjoyment Scale
- **Why It Measures the Construct:** Measures debugging logic, deductive reasoning, and computational troubleshooting⁹.
- **Scoring Direction:** Positive ($1 \rightarrow 5$)
- **Potential Ambiguity:** Assumes mobile app exposure.
- **Cultural/Context Concerns:** Widespread smartphone familiarity among adolescents in AP/Telangana⁶¹.
- **Reading Difficulty Estimate:** FKGL 5.0
- **Class 10 Suitability:** Fully suitable.
- **Pilot Requirement:** Test in pilot for socioeconomic variation.

Item QCR-09

- **Construct Measured:** Quantitative & Computational Reasoning (QCR)
- **Question:** How much do you enjoy solving logic puzzles where you must deduce who is sitting where based on a list of clues?
- **Answer Choices:** (1) Strongly Dislike (2) Dislike (3) Neutral / Unsure (4) Like (5) Strongly Like
- **Question Type:** 5-point Likert Enjoyment Scale
- **Why It Measures the Construct:** Evaluates symbolic deduction and constraint-satisfaction logic⁹.
- **Scoring Direction:** Positive ($1 \rightarrow 5$)
- **Potential Ambiguity:** None.

- **Cultural/Context Concerns:** Common format in regional talent search exams²².
- **Reading Difficulty Estimate:** FKGL 4.6
- **Class 10 Suitability:** Fully suitable.
- **Pilot Requirement:** Include in candidate reserve pool.

Item QCR-10

- **Construct Measured:** Quantitative & Computational Reasoning (QCR)
- **Question:** How much do you enjoy checking long calculation sheets to find and fix a single numerical mistake?
- **Answer Choices:** (1) Strongly Dislike (2) Dislike (3) Neutral / Unsure (4) Like (5) Strongly Like
- **Question Type:** 5-point Likert Enjoyment Scale
- **Why It Measures the Construct:** Evaluates numerical accuracy and tolerance for detailed auditing³.
- **Scoring Direction:** Positive ($1 \rightarrow 5$)
- **Potential Ambiguity:** None.
- **Cultural/Context Concerns:** Connects mathematics with accounting precision (MPC/MEC overlap)³.
- **Reading Difficulty Estimate:** FKGL 4.4
- **Class 10 Suitability:** Fully suitable.
- **Pilot Requirement:** Include in candidate reserve pool.

Technical & Mechanical Design (TMD)

Item TMD-01

- **Construct Measured:** Technical & Mechanical Design (TMD)
- **Question:** How much do you enjoy taking apart a broken mechanical object (like a bicycle bell, toy, or pen) to see how its gears and springs work?
- **Answer Choices:** (1) Strongly Dislike (2) Dislike (3) Neutral / Unsure (4) Like (5) Strongly Like
- **Question Type:** 5-point Likert Enjoyment Scale
- **Why It Measures the Construct:** Measures mechanical curiosity and physical mechanism deconstruction¹.
- **Scoring Direction:** Positive ($1 \rightarrow 5$)
- **Potential Ambiguity:** None.
- **Cultural/Context Concerns:** Uses universally accessible objects, avoiding assumptions of expensive tools⁵³.
- **Reading Difficulty Estimate:** FKGL 4.5
- **Class 10 Suitability:** Fully suitable.
- **Pilot Requirement:** Include in alpha pilot.

Item TMD-02

- **Construct Measured:** Technical & Mechanical Design (TMD)
- **Question:** How much do you enjoy drawing scale floor plans or diagrams of how a

house, room, or bridge should be constructed?

- **Answer Choices:** (1) Strongly Dislike (2) Dislike (3) Neutral / Unsure (4) Like (5) Strongly Like
- **Question Type:** 5-point Likert Enjoyment Scale
- **Why It Measures the Construct:** Assesses spatial drafting, architectural visualization, and structural design affinity⁹.
- **Scoring Direction:** Positive ($1 \rightarrow 5$)
- **Potential Ambiguity:** None.
- **Cultural/Context Concerns:** Paper-and-pencil task; highly relevant for Civil Engineering and Architecture¹.
- **Reading Difficulty Estimate:** FKGL 4.7
- **Class 10 Suitability:** Fully suitable.
- **Pilot Requirement:** Include in alpha pilot.

Item TMD-03

- **Construct Measured:** Technical & Mechanical Design (TMD)
- **Question:** How much do you enjoy connecting batteries, wires, small bulbs, and switches to build working electrical circuits?
- **Answer Choices:** (1) Strongly Dislike (2) Dislike (3) Neutral / Unsure (4) Like (5) Strongly Like
- **Question Type:** 5-point Likert Enjoyment Scale
- **Why It Measures the Construct:** Evaluates affinity for electrical hardware and hands-on physical science application⁹.
- **Scoring Direction:** Positive ($1 \rightarrow 5$)
- **Potential Ambiguity:** None.
- **Cultural/Context Concerns:** Aligned with the Class 10 Physics unit on Electric Current¹⁸.
- **Reading Difficulty Estimate:** FKGL 4.8
- **Class 10 Suitability:** Fully suitable.
- **Pilot Requirement:** Include in alpha pilot.

Item TMD-04

- **Construct Measured:** Technical & Mechanical Design (TMD)
- **Question:** How much do you enjoy using hand tools (like screwdrivers, pliers, or hammers) to repair a loose hinge or fix furniture at home?
- **Answer Choices:** (1) Strongly Dislike (2) Dislike (3) Neutral / Unsure (4) Like (5) Strongly Like
- **Question Type:** 5-point Likert Enjoyment Scale
- **Why It Measures the Construct:** Captures hands-on repair and tool engagement (Holland Realistic)⁵.
- **Scoring Direction:** Positive ($1 \rightarrow 5$)
- **Potential Ambiguity:** None.
- **Cultural/Context Concerns:** Framed around routine domestic repairs to avoid

caste/class stigma regarding manual labor⁷.

- **Reading Difficulty Estimate:** FKGL 4.4
- **Class 10 Suitability:** Fully suitable.
- **Pilot Requirement:** Include in alpha pilot.

Item TMD-05

- **Construct Measured:** Technical & Mechanical Design (TMD)
- **Question:** How much do you enjoy reading instruction manuals or diagrams that explain how a new electronic machine works?
- **Answer Choices:** (1) Strongly Dislike (2) Dislike (3) Neutral / Unsure (4) Like (5) Strongly Like
- **Question Type:** 5-point Likert Enjoyment Scale
- **Why It Measures the Construct:** Assesses technical literacy and interest in systematic equipment operations⁹.
- **Scoring Direction:** Positive (1 → 5)
- **Potential Ambiguity:** None.
- **Cultural/Context Concerns:** Familiar activity across domestic appliances (mixers, pumps, TVs)⁵³.
- **Reading Difficulty Estimate:** FKGL 5.1
- **Class 10 Suitability:** Fully suitable.
- **Pilot Requirement:** Include in alpha pilot.

Item TMD-06

- **Construct Measured:** Technical & Mechanical Design (TMD)
- **Question:** How much do you enjoy building sturdy, three-dimensional models (like towers, vehicles, or cardboard bridges) using basic craft materials?
- **Answer Choices:** (1) Strongly Dislike (2) Dislike (3) Neutral / Unsure (4) Like (5) Strongly Like
- **Question Type:** 5-point Likert Enjoyment Scale
- **Why It Measures the Construct:** Measures physical fabrication interest and applied spatial problem-solving¹.
- **Scoring Direction:** Positive (1 → 5)
- **Potential Ambiguity:** None.
- **Cultural/Context Concerns:** Familiar school project activity requiring only basic cardboard and glue⁵³.
- **Reading Difficulty Estimate:** FKGL 4.9
- **Class 10 Suitability:** Fully suitable.
- **Pilot Requirement:** Include in alpha pilot.

Item TMD-07

- **Construct Measured:** Technical & Mechanical Design (TMD)
- **Question:** How much do you enjoy learning how automobile engines, brakes, and gears transfer power to make a vehicle move?
- **Answer Choices:** (1) Strongly Dislike (2) Dislike (3) Neutral / Unsure (4) Like (5) Strongly

Like

- **Question Type:** 5-point Likert Enjoyment Scale
- **Why It Measures the Construct:** Evaluates mechanical engineering affinity and vehicular systems interest¹.
- **Scoring Direction:** Positive ($1 \rightarrow 5$)
- **Potential Ambiguity:** May attract superficial interest in automobiles.
- **Cultural/Context Concerns:** Monitor for potential gender skew during validation⁶.
- **Reading Difficulty Estimate:** FKGL 4.8
- **Class 10 Suitability:** Fully suitable.
- **Pilot Requirement:** Test in pilot for gender DIF.

Item TMD-08

- **Construct Measured:** Technical & Mechanical Design (TMD)
- **Question:** How much do you enjoy installing software updates, setting up internet routers, or troubleshooting tech hardware connections?
- **Answer Choices:** (1) Strongly Dislike (2) Dislike (3) Neutral / Unsure (4) Like (5) Strongly Like
- **Question Type:** 5-point Likert Enjoyment Scale
- **Why It Measures the Construct:** Measures IT hardware administration and practical technical troubleshooting¹.
- **Scoring Direction:** Positive ($1 \rightarrow 5$)
- **Potential Ambiguity:** Rural students may lack direct home broadband router experience.
- **Cultural/Context Concerns:** Focus phrasing on phone setup and Wi-Fi connections².
- **Reading Difficulty Estimate:** FKGL 5.3
- **Class 10 Suitability:** Suitable with slight refinement.
- **Pilot Requirement:** Test in pilot across urban and rural cohorts.

Item TMD-09

- **Construct Measured:** Technical & Mechanical Design (TMD)
- **Question:** How much do you enjoy designing physical safety mechanisms, such as protective covers or guards, to make tools safer to operate?
- **Answer Choices:** (1) Strongly Dislike (2) Dislike (3) Neutral / Unsure (4) Like (5) Strongly Like
- **Question Type:** 5-point Likert Enjoyment Scale
- **Why It Measures the Construct:** Assesses industrial design intuition, ergonomics, and practical problem-solving¹.
- **Scoring Direction:** Positive ($1 \rightarrow 5$)
- **Potential Ambiguity:** None.
- **Cultural/Context Concerns:** Clear, practical concept.
- **Reading Difficulty Estimate:** FKGL 5.0
- **Class 10 Suitability:** Fully suitable.
- **Pilot Requirement:** Include in candidate reserve pool.

Item TMD-10

- **Construct Measured:** Technical & Mechanical Design (TMD)
- **Question:** How much do you enjoy measuring physical dimensions with a steel ruler or tape to verify that wooden or metal parts fit together perfectly?
- **Answer Choices:** (1) Strongly Dislike (2) Dislike (3) Neutral / Unsure (4) Like (5) Strongly Like
- **Question Type:** 5-point Likert Enjoyment Scale
- **Why It Measures the Construct:** Evaluates spatial precision, dimensional tolerance, and fabrication interest⁹.
- **Scoring Direction:** Positive ($1 \rightarrow 5$)
- **Potential Ambiguity:** None.
- **Cultural/Context Concerns:** Grounded in practical construction tasks.
- **Reading Difficulty Estimate:** FKGL 4.7
- **Class 10 Suitability:** Fully suitable.
- **Pilot Requirement:** Include in candidate reserve pool.

Commercial & Economic Enterprise (CEE)

Item CEE-01

- **Construct Measured:** Commercial & Economic Enterprise (CEE)
- **Question:** How much do you enjoy calculating profit, loss, and discounts when buying items in bulk or planning a sale?
- **Answer Choices:** (1) Strongly Dislike (2) Dislike (3) Neutral / Unsure (4) Like (5) Strongly Like
- **Question Type:** 5-point Likert Enjoyment Scale
- **Why It Measures the Construct:** Measures foundational commercial arithmetic and business acumen, aligned with Commerce and Accountancy³.
- **Scoring Direction:** Positive ($1 \rightarrow 5$)
- **Potential Ambiguity:** None.
- **Cultural/Context Concerns:** Part of secondary school mathematics and relevant to retail trade¹.
- **Reading Difficulty Estimate:** FKGL 4.3
- **Class 10 Suitability:** Fully suitable.
- **Pilot Requirement:** Include in alpha pilot.

Item CEE-02

- **Construct Measured:** Commercial & Economic Enterprise (CEE)
- **Question:** How much do you enjoy learning how local markets determine the daily prices of vegetables, petrol, gold, or grains?
- **Answer Choices:** (1) Strongly Dislike (2) Dislike (3) Neutral / Unsure (4) Like (5) Strongly Like
- **Question Type:** 5-point Likert Enjoyment Scale
- **Why It Measures the Construct:** Assesses interest in microeconomics, price theory, and

market dynamics (MEC/CEC)³.

- **Scoring Direction:** Positive ($1 \rightarrow 5$)
- **Potential Ambiguity:** None.
- **Cultural/Context Concerns:** Taught in Class 10 Social Studies (Production and Employment / Rampur Village Economy)¹⁸.
- **Reading Difficulty Estimate:** FKGL 4.9
- **Class 10 Suitability:** Fully suitable.
- **Pilot Requirement:** Include in alpha pilot.

Item CEE-03

- **Construct Measured:** Commercial & Economic Enterprise (CEE)
- **Question:** How much do you enjoy planning how to advertise and pitch a new product to convince classmates to buy it for a school fundraiser?
- **Answer Choices:** (1) Strongly Dislike (2) Dislike (3) Neutral / Unsure (4) Like (5) Strongly Like
- **Question Type:** 5-point Likert Enjoyment Scale
- **Why It Measures the Construct:** Measures marketing, persuasive communication, and entrepreneurial enterprise (Holland Enterprising)⁵.
- **Scoring Direction:** Positive ($1 \rightarrow 5$)
- **Potential Ambiguity:** None.
- **Cultural/Context Concerns:** Framed around an accessible school event rather than high-pressure corporate sales¹⁹.
- **Reading Difficulty Estimate:** FKGL 5.1
- **Class 10 Suitability:** Fully suitable.
- **Pilot Requirement:** Include in alpha pilot.

Item CEE-04

- **Construct Measured:** Commercial & Economic Enterprise (CEE)
- **Question:** How much do you enjoy maintaining a neat, accurate logbook of income, pocket money, and expenses to track your savings?
- **Answer Choices:** (1) Strongly Dislike (2) Dislike (3) Neutral / Unsure (4) Like (5) Strongly Like
- **Question Type:** 5-point Likert Enjoyment Scale
- **Why It Measures the Construct:** Evaluates bookkeeping discipline, budgetary habits, and accounting orientation³.
- **Scoring Direction:** Positive ($1 \rightarrow 5$)
- **Potential Ambiguity:** None.
- **Cultural/Context Concerns:** Grounded in personal financial habits; avoids assuming high family income⁵³.
- **Reading Difficulty Estimate:** FKGL 4.6
- **Class 10 Suitability:** Fully suitable.
- **Pilot Requirement:** Include in alpha pilot.

Item CEE-05

- **Construct Measured:** Commercial & Economic Enterprise (CEE)
- **Question:** How much do you enjoy reading or discussing news stories about new business startups, banks, and economic growth in India?
- **Answer Choices:** (1) Strongly Dislike (2) Dislike (3) Neutral / Unsure (4) Like (5) Strongly Like
- **Question Type:** 5-point Likert Enjoyment Scale
- **Why It Measures the Construct:** Captures commercial awareness, economic curiosity, and business interest¹.
- **Scoring Direction:** Positive ($1 \rightarrow 5$)
- **Potential Ambiguity:** None.
- **Cultural/Context Concerns:** Prominent in regional news and media outlets¹⁸.
- **Reading Difficulty Estimate:** FKGL 5.0
- **Class 10 Suitability:** Fully suitable.
- **Pilot Requirement:** Include in alpha pilot.

Item CEE-06

- **Construct Measured:** Commercial & Economic Enterprise (CEE)
- **Question:** How much do you enjoy negotiating prices or terms with someone to reach a deal where both sides feel satisfied?
- **Answer Choices:** (1) Strongly Dislike (2) Dislike (3) Neutral / Unsure (4) Like (5) Strongly Like
- **Question Type:** 5-point Likert Enjoyment Scale
- **Why It Measures the Construct:** Assesses commercial negotiation, trade psychology, and transactional communication⁹.
- **Scoring Direction:** Positive ($1 \rightarrow 5$)
- **Potential Ambiguity:** None.
- **Cultural/Context Concerns:** Common everyday experience in local markets.
- **Reading Difficulty Estimate:** FKGL 4.8
- **Class 10 Suitability:** Fully suitable.
- **Pilot Requirement:** Include in alpha pilot.

Item CEE-07

- **Construct Measured:** Commercial & Economic Enterprise (CEE)
- **Question:** How much do you enjoy figuring out how banks earn interest and manage the savings of their account holders?
- **Answer Choices:** (1) Strongly Dislike (2) Dislike (3) Neutral / Unsure (4) Like (5) Strongly Like
- **Question Type:** 5-point Likert Enjoyment Scale
- **Why It Measures the Construct:** Evaluates interest in banking operations, financial intermediation, and interest calculations¹.
- **Scoring Direction:** Positive ($1 \rightarrow 5$)

- **Potential Ambiguity:** None.
- **Cultural/Context Concerns:** Connects with Class 10 Mathematics (Simple and Compound Interest) and Social Studies¹⁸.
- **Reading Difficulty Estimate:** FKGL 4.7
- **Class 10 Suitability:** Fully suitable.
- **Pilot Requirement:** Include in candidate reserve pool.

Item CEE-08

- **Construct Measured:** Commercial & Economic Enterprise (CEE)
- **Question:** How much do you enjoy brainstorming ideas for a small shop or enterprise that could provide a needed service in your neighborhood?
- **Answer Choices:** (1) Strongly Dislike (2) Dislike (3) Neutral / Unsure (4) Like (5) Strongly Like
- **Question Type:** 5-point Likert Enjoyment Scale
- **Why It Measures the Construct:** Measures entrepreneurial initiative, localized market opportunity analysis, and enterprise design¹.
- **Scoring Direction:** Positive ($1 \rightarrow 5$)
- **Potential Ambiguity:** None.
- **Cultural/Context Concerns:** Relatable across rural mandals and urban centers².
- **Reading Difficulty Estimate:** FKGL 5.1
- **Class 10 Suitability:** Fully suitable.
- **Pilot Requirement:** Include in candidate reserve pool.

Item CEE-09

- **Construct Measured:** Commercial & Economic Enterprise (CEE)
- **Question:** How much do you enjoy comparing the delivery charges, fees, and product prices across different shopping apps or local stores to find the best deal?
- **Answer Choices:** (1) Strongly Dislike (2) Dislike (3) Neutral / Unsure (4) Like (5) Strongly Like
- **Question Type:** 5-point Likert Enjoyment Scale
- **Why It Measures the Construct:** Assesses consumer economics, price comparison habits, and practical market analytics⁹.
- **Scoring Direction:** Positive ($1 \rightarrow 5$)
- **Potential Ambiguity:** None.
- **Cultural/Context Concerns:** Common activity in mobile-enabled households across AP and Telangana.
- **Reading Difficulty Estimate:** FKGL 5.0
- **Class 10 Suitability:** Fully suitable.
- **Pilot Requirement:** Test in pilot for rural/urban variation.

Item CEE-10

- **Construct Measured:** Commercial & Economic Enterprise (CEE)
- **Question:** How much do you enjoy managing and distributing supplies or funds for a sports tournament or school annual day?

- **Answer Choices:** (1) Strongly Dislike (2) Dislike (3) Neutral / Unsure (4) Like (5) Strongly Like
- **Question Type:** 5-point Likert Enjoyment Scale
- **Why It Measures the Construct:** Evaluates resource allocation, budget stewardship, and administrative management³.
- **Scoring Direction:** Positive ($1 \rightarrow 5$)
- **Potential Ambiguity:** None.
- **Cultural/Context Concerns:** Routine co-curricular responsibility in Indian secondary schools¹⁸.
- **Reading Difficulty Estimate:** FKGL 4.8
- **Class 10 Suitability:** Fully suitable.
- **Pilot Requirement:** Include in candidate reserve pool.

Social, Helping & Administrative Coordination (SHC)

Item SHC-01

- **Construct Measured:** Social, Helping & Administrative Coordination (SHC)
- **Question:** How much do you enjoy patiently explaining a difficult homework topic to a classmate who is struggling to understand it?
- **Answer Choices:** (1) Strongly Dislike (2) Dislike (3) Neutral / Unsure (4) Like (5) Strongly Like
- **Question Type:** 5-point Likert Enjoyment Scale
- **Why It Measures the Construct:** Measures instructional affinity, empathetic teaching, and human development orientation (Holland Social)⁵.
- **Scoring Direction:** Positive ($1 \rightarrow 5$)
- **Potential Ambiguity:** None.
- **Cultural/Context Concerns:** Universal school peer interaction¹⁸.
- **Reading Difficulty Estimate:** FKGL 4.5
- **Class 10 Suitability:** Fully suitable.
- **Pilot Requirement:** Include in alpha pilot.

Item SHC-02

- **Construct Measured:** Social, Helping & Administrative Coordination (SHC)
- **Question:** When friends have an argument or disagreement, how much do you enjoy stepping in to help them find a peaceful, fair solution?
- **Answer Choices:** (1) Strongly Dislike (2) Dislike (3) Neutral / Unsure (4) Like (5) Strongly Like
- **Question Type:** 5-point Likert Enjoyment Scale
- **Why It Measures the Construct:** Assesses interpersonal mediation, conflict resolution, and social harmony orientation⁹.
- **Scoring Direction:** Positive ($1 \rightarrow 5$)
- **Potential Ambiguity:** None.
- **Cultural/Context Concerns:** Natural adolescent peer group experience⁶.

- **Reading Difficulty Estimate:** FKGL 4.7
- **Class 10 Suitability:** Fully suitable.
- **Pilot Requirement:** Include in alpha pilot.

Item SHC-03

- **Construct Measured:** Social, Helping & Administrative Coordination (SHC)
- **Question:** How much do you enjoy learning about government welfare programs and how citizens can access their legal rights and benefits?
- **Answer Choices:** (1) Strongly Dislike (2) Dislike (3) Neutral / Unsure (4) Like (5) Strongly Like
- **Question Type:** 5-point Likert Enjoyment Scale
- **Why It Measures the Construct:** Evaluates civic awareness, public administration interest, and legal advocacy, core to Civics in CEC³.
- **Scoring Direction:** Positive ($1 \rightarrow 5$)
- **Potential Ambiguity:** None.
- **Cultural/Context Concerns:** Highly prominent in regional social discourse and school curricula¹⁸.
- **Reading Difficulty Estimate:** FKGL 5.2
- **Class 10 Suitability:** Fully suitable.
- **Pilot Requirement:** Include in alpha pilot.

Item SHC-04

- **Construct Measured:** Social, Helping & Administrative Coordination (SHC)
- **Question:** How much do you enjoy volunteering your time to help organize health camps, blood donation drives, or neighborhood cleanups?
- **Answer Choices:** (1) Strongly Dislike (2) Dislike (3) Neutral / Unsure (4) Like (5) Strongly Like
- **Question Type:** 5-point Likert Enjoyment Scale
- **Why It Measures the Construct:** Assesses community service commitment, public health orientation, and civic engagement⁹.
- **Scoring Direction:** Positive ($1 \rightarrow 5$)
- **Potential Ambiguity:** None.
- **Cultural/Context Concerns:** Familiar community and school activities (NSS, Scouts, local welfare groups)².
- **Reading Difficulty Estimate:** FKGL 5.0
- **Class 10 Suitability:** Fully suitable.
- **Pilot Requirement:** Include in alpha pilot.

Item SHC-05

- **Construct Measured:** Social, Helping & Administrative Coordination (SHC)
- **Question:** When someone in your family or circle feels sad, how much do you enjoy listening to their personal worries and comforting them?
- **Answer Choices:** (1) Strongly Dislike (2) Dislike (3) Neutral / Unsure (4) Like (5) Strongly Like

- **Question Type:** 5-point Likert Enjoyment Scale
- **Why It Measures the Construct:** Measures counseling empathy, emotional support orientation, and interpersonal caregiving⁹.
- **Scoring Direction:** Positive (**1 → 5**)
- **Potential Ambiguity:** None.
- **Cultural/Context Concerns:** Monitor for potential gender response differences⁶.
- **Reading Difficulty Estimate:** FKGL 4.2
- **Class 10 Suitability:** Fully suitable.
- **Pilot Requirement:** Include in alpha pilot.

Item SHC-06

- **Construct Measured:** Social, Helping & Administrative Coordination (SHC)
- **Question:** How much do you enjoy learning how a district court or village panchayat hears evidence to make fair legal judgments?
- **Answer Choices:** (1) Strongly Dislike (2) Dislike (3) Neutral / Unsure (4) Like (5) Strongly Like
- **Question Type:** 5-point Likert Enjoyment Scale
- **Why It Measures the Construct:** Assesses interest in legal systems, judicial procedures, and institutional governance (CEC Civics/Law)¹.
- **Scoring Direction:** Positive (**1 → 5**)
- **Potential Ambiguity:** None.
- **Cultural/Context Concerns:** Aligned with Class 10 Social Studies (Indian Constitution, Election Process, Local Governance)¹⁸.
- **Reading Difficulty Estimate:** FKGL 5.1
- **Class 10 Suitability:** Fully suitable.
- **Pilot Requirement:** Include in alpha pilot.

Item SHC-07

- **Construct Measured:** Social, Helping & Administrative Coordination (SHC)
- **Question:** How much do you enjoy organizing rules, team schedules, and duties to make sure a student club runs smoothly?
- **Answer Choices:** (1) Strongly Dislike (2) Dislike (3) Neutral / Unsure (4) Like (5) Strongly Like
- **Question Type:** 5-point Likert Enjoyment Scale
- **Why It Measures the Construct:** Evaluates administrative coordination, procedural order, and group governance⁹.
- **Scoring Direction:** Positive (**1 → 5**)
- **Potential Ambiguity:** None.
- **Cultural/Context Concerns:** Accessible co-curricular framing¹⁸.
- **Reading Difficulty Estimate:** FKGL 4.8
- **Class 10 Suitability:** Fully suitable.
- **Pilot Requirement:** Include in candidate reserve pool.

Item SHC-08

- **Construct Measured:** Social, Helping & Administrative Coordination (SHC)
- **Question:** How much do you enjoy assisting elderly people or hospital patients with their daily medicine routines and personal comfort?
- **Answer Choices:** (1) Strongly Dislike (2) Dislike (3) Neutral / Unsure (4) Like (5) Strongly Like
- **Question Type:** 5-point Likert Enjoyment Scale
- **Why It Measures the Construct:** Evaluates bedside patient care, clinical nursing affinity, and healthcare empathy (BiPC)⁹.
- **Scoring Direction:** Positive ($1 \rightarrow 5$)
- **Potential Ambiguity:** None.
- **Cultural/Context Concerns:** Highly respected behavioral tradition in multi-generational families⁷.
- **Reading Difficulty Estimate:** FKGL 4.9
- **Class 10 Suitability:** Fully suitable.
- **Pilot Requirement:** Include in candidate reserve pool.

Item SHC-09

- **Construct Measured:** Social, Helping & Administrative Coordination (SHC)
- **Question:** How much do you enjoy participating in classroom discussions about historical leaders who fought for equal human rights and social justice?
- **Answer Choices:** (1) Strongly Dislike (2) Dislike (3) Neutral / Unsure (4) Like (5) Strongly Like
- **Question Type:** 5-point Likert Enjoyment Scale
- **Why It Measures the Construct:** Taps sociopolitical consciousness, historical empathy, and public reform interest (CEC/HEC)¹.
- **Scoring Direction:** Positive ($1 \rightarrow 5$)
- **Potential Ambiguity:** None.
- **Cultural/Context Concerns:** Direct connection with Class 10 curriculum units on Social Movements and Human Rights¹⁸.
- **Reading Difficulty Estimate:** FKGL 5.4
- **Class 10 Suitability:** Fully suitable.
- **Pilot Requirement:** Include in candidate reserve pool.

Item SHC-10

- **Construct Measured:** Social, Helping & Administrative Coordination (SHC)
- **Question:** How much do you enjoy collecting feedback and survey opinions from classmates to find out how to improve school facilities?
- **Answer Choices:** (1) Strongly Dislike (2) Dislike (3) Neutral / Unsure (4) Like (5) Strongly Like
- **Question Type:** 5-point Likert Enjoyment Scale
- **Why It Measures the Construct:** Assesses community consultation, social survey interest, and institutional problem solving⁹.

- **Scoring Direction:** Positive ($1 \rightarrow 5$)
- **Potential Ambiguity:** None.
- **Cultural/Context Concerns:** Straightforward school-based scenario².
- **Reading Difficulty Estimate:** FKGL 5.0
- **Class 10 Suitability:** Fully suitable.
- **Pilot Requirement:** Include in candidate reserve pool.

Creative & Expressive Arts (CEA)

Item CEA-01

- **Construct Measured:** Creative & Expressive Arts (CEA)
- **Question:** How much do you enjoy writing imaginative poems, short stories, or personal diary entries to express your feelings?
- **Answer Choices:** (1) Strongly Dislike (2) Dislike (3) Neutral / Unsure (4) Like (5) Strongly Like
- **Question Type:** 5-point Likert Enjoyment Scale
- **Why It Measures the Construct:** Assesses literary creativity, verbal expressiveness, and narrative imagination (Holland Artistic)⁵.
- **Scoring Direction:** Positive ($1 \rightarrow 5$)
- **Potential Ambiguity:** None.
- **Cultural/Context Concerns:** Direct alignment with Class 10 English and Telugu creative writing activities⁴².
- **Reading Difficulty Estimate:** FKGL 4.5
- **Class 10 Suitability:** Fully suitable.
- **Pilot Requirement:** Include in alpha pilot.

Item CEA-02

- **Construct Measured:** Creative & Expressive Arts (CEA)
- **Question:** How much do you enjoy sketching pencil portraits, landscape drawings, or digital poster illustrations?
- **Answer Choices:** (1) Strongly Dislike (2) Dislike (3) Neutral / Unsure (4) Like (5) Strongly Like
- **Question Type:** 5-point Likert Enjoyment Scale
- **Why It Measures the Construct:** Measures visual arts affinity, graphic composition, and visual-spatial creativity¹.
- **Scoring Direction:** Positive ($1 \rightarrow 5$)
- **Potential Ambiguity:** None.
- **Cultural/Context Concerns:** Encompasses both traditional pencil drawing and mobile design tools⁵³.
- **Reading Difficulty Estimate:** FKGL 4.6
- **Class 10 Suitability:** Fully suitable.
- **Pilot Requirement:** Include in alpha pilot.

Item CEA-03

- **Construct Measured:** Creative & Expressive Arts (CEA)
- **Question:** How much do you enjoy selecting colors, fonts, and photo layouts to make an invitation card or banner look attractive?
- **Answer Choices:** (1) Strongly Dislike (2) Dislike (3) Neutral / Unsure (4) Like (5) Strongly Like
- **Question Type:** 5-point Likert Enjoyment Scale
- **Why It Measures the Construct:** Evaluates graphic design, aesthetic sensibility, and visual communication¹.
- **Scoring Direction:** Positive ($1 \rightarrow 5$)
- **Potential Ambiguity:** None.
- **Cultural/Context Concerns:** Common task for school functions, community events, and family celebrations¹⁸.
- **Reading Difficulty Estimate:** FKGL 4.8
- **Class 10 Suitability:** Fully suitable.
- **Pilot Requirement:** Include in alpha pilot.

Item CEA-04

- **Construct Measured:** Creative & Expressive Arts (CEA)
- **Question:** How much do you enjoy performing in school plays, acting out characters, or reciting dramatic dialogues on stage?
- **Answer Choices:** (1) Strongly Dislike (2) Dislike (3) Neutral / Unsure (4) Like (5) Strongly Like
- **Question Type:** 5-point Likert Enjoyment Scale
- **Why It Measures the Construct:** Measures theatrical performance, bodily-kinesthetic expression, and dramatic interpretation⁹.
- **Scoring Direction:** Positive ($1 \rightarrow 5$)
- **Potential Ambiguity:** Shy or introverted creative students may rate this low; balanced by written and visual design items⁵.
- **Cultural/Context Concerns:** Features regularly in regional school annual days and cultural celebrations⁶¹.
- **Reading Difficulty Estimate:** FKGL 4.9
- **Class 10 Suitability:** Fully suitable.
- **Pilot Requirement:** Include in alpha pilot.

Item CEA-05

- **Construct Measured:** Creative & Expressive Arts (CEA)
- **Question:** How much do you enjoy analyzing movies or web series to figure out how camera angles, background music, and storytelling create emotion?
- **Answer Choices:** (1) Strongly Dislike (2) Dislike (3) Neutral / Unsure (4) Like (5) Strongly Like
- **Question Type:** 5-point Likert Enjoyment Scale
- **Why It Measures the Construct:** Assesses critical media analysis, cinematic narrative theory, and audiovisual appreciation¹.

- **Scoring Direction:** Positive ($1 \rightarrow 5$)
- **Potential Ambiguity:** Phrased to evaluate analytical creative evaluation rather than passive viewing⁴².
- **Cultural/Context Concerns:** Connects directly with Class 10 English Unit 4 (*Films and Theatre*)¹⁸.
- **Reading Difficulty Estimate:** FKGL 5.3
- **Class 10 Suitability:** Fully suitable.
- **Pilot Requirement:** Include in alpha pilot.

Item CEA-06

- **Construct Measured:** Creative & Expressive Arts (CEA)
- **Question:** How much do you enjoy practicing musical instruments, humming melodies, or arranging rhythms and beats?
- **Answer Choices:** (1) Strongly Dislike (2) Dislike (3) Neutral / Unsure (4) Like (5) Strongly Like
- **Question Type:** 5-point Likert Enjoyment Scale
- **Why It Measures the Construct:** Measures musical interest, acoustic sensitivity, and melodic creativity⁹.
- **Scoring Direction:** Positive ($1 \rightarrow 5$)
- **Potential Ambiguity:** None.
- **Cultural/Context Concerns:** Encompasses formal instruments as well as informal singing and rhythmic patterns⁵³.
- **Reading Difficulty Estimate:** FKGL 4.7
- **Class 10 Suitability:** Fully suitable.
- **Pilot Requirement:** Include in alpha pilot.

Item CEA-07

- **Construct Measured:** Creative & Expressive Arts (CEA)
- **Question:** How much do you enjoy designing unique clothing styles, festive costumes, or matching decorative fabric patterns?
- **Answer Choices:** (1) Strongly Dislike (2) Dislike (3) Neutral / Unsure (4) Like (5) Strongly Like
- **Question Type:** 5-point Likert Enjoyment Scale
- **Why It Measures the Construct:** Assesses fashion design, textile aesthetics, and decorative visual styling¹.
- **Scoring Direction:** Positive ($1 \rightarrow 5$)
- **Potential Ambiguity:** None.
- **Cultural/Context Concerns:** Connects with rich regional textile traditions (e.g., Pochampally, Mangalagiri).
- **Reading Difficulty Estimate:** FKGL 4.8
- **Class 10 Suitability:** Fully suitable.
- **Pilot Requirement:** Include in candidate reserve pool.

Item CEA-08

- **Construct Measured:** Creative & Expressive Arts (CEA)
- **Question:** How much do you enjoy editing recorded video clips on a phone using sound effects, transitions, and text animations?
- **Answer Choices:** (1) Strongly Dislike (2) Dislike (3) Neutral / Unsure (4) Like (5) Strongly Like
- **Question Type:** 5-point Likert Enjoyment Scale
- **Why It Measures the Construct:** Evaluates digital multimedia editing, contemporary creative expression, and video production¹.
- **Scoring Direction:** Positive ($1 \rightarrow 5$)
- **Potential Ambiguity:** None.
- **Cultural/Context Concerns:** Highly familiar activity among adolescents using mobile devices⁶¹.
- **Reading Difficulty Estimate:** FKGL 5.0
- **Class 10 Suitability:** Fully suitable.
- **Pilot Requirement:** Include in candidate reserve pool.

Item CEA-09

- **Construct Measured:** Creative & Expressive Arts (CEA)
- **Question:** How much do you enjoy writing honest reviews or opinions about newly published books, essays, or theater plays?
- **Answer Choices:** (1) Strongly Dislike (2) Dislike (3) Neutral / Unsure (4) Like (5) Strongly Like
- **Question Type:** 5-point Likert Enjoyment Scale
- **Why It Measures the Construct:** Evaluates literary analysis, critical commentary, and formal written discourse⁹.
- **Scoring Direction:** Positive ($1 \rightarrow 5$)
- **Potential Ambiguity:** None.
- **Cultural/Context Concerns:** Taught in secondary language curriculum units on review writing⁴².
- **Reading Difficulty Estimate:** FKGL 5.2
- **Class 10 Suitability:** Fully suitable.
- **Pilot Requirement:** Include in candidate reserve pool.

Item CEA-10

- **Construct Measured:** Creative & Expressive Arts (CEA)
- **Question:** How much do you enjoy rearranging room furniture, wall decorations, and lighting to create a pleasant living atmosphere?
- **Answer Choices:** (1) Strongly Dislike (2) Dislike (3) Neutral / Unsure (4) Like (5) Strongly Like
- **Question Type:** 5-point Likert Enjoyment Scale
- **Why It Measures the Construct:** Assesses interior spatial aesthetics, environmental styling, and decorative design⁹.

- **Scoring Direction:** Positive ($1 \rightarrow 5$)
- **Potential Ambiguity:** None.
- **Cultural/Context Concerns:** Familiar domestic activity across socioeconomic backgrounds⁵³.
- **Reading Difficulty Estimate:** FKGL 5.0
- **Class 10 Suitability:** Fully suitable.
- **Pilot Requirement:** Include in candidate reserve pool.

Unscored Contextual & Filtering Module (CTX)

Item CTX-01: Baseline Subject Ease

- **Construct Measured:** Academic Self-Concept / Subject Ease (Unscored Context)
- **Question:** When studying for exams in Class 10, which subject area generally felt easiest and most natural for you to understand?
- **Answer Choices:**
 - (A) Mathematics (Algebra, Geometry, Trigonometry)
 - (B) Physical Science (Physics, Chemistry laws and formulas)
 - (C) Biological Science (Plants, Animals, Human Body)
 - (D) Social Studies (History, Economics, Civics, Geography)
 - (E) Languages and Literature (Telugu, English, Hindi)
- **Question Type:** Single-select categorical
- **Why It Measures the Construct:** Provides self-reported academic ease to cross-reference against interest profiles, identifying interest-ability discrepancies⁸.
- **Scoring Direction:** Unscored (Contextual Diagnostic Reference).
- **Potential Ambiguity:** None.
- **Cultural/Context Concerns:** Matches the standard SSC subject examination structure³⁹.
- **Reading Difficulty Estimate:** FKGL 4.8
- **Class 10 Suitability:** Fully suitable.
- **Pilot Requirement:** Include in alpha pilot.

Item CTX-02: Familial Career Expectations

- **Construct Measured:** Familial Socio-Cognitive Expectations (Unscored Context)
- **Question:** What stream or career path do your parents or family elders most hope you will choose after Class 10?
- **Answer Choices:**
 - (A) Engineering / Technology / Computing (MPC path)
 - (B) Medical / Healthcare / Agriculture (BiPC path)
 - (C) Commerce / Chartered Accountancy / Business (MEC / CEC path)
 - (D) Civil Services / Government Officer / Law (UPSC / State Exams)
 - (E) They encourage me to explore whatever matches my personal interests
 - (F) We have not discussed this yet
- **Question Type:** Single-select categorical
- **Why It Measures the Construct:** Identifies family expectations and livelihood beliefs as defined by Arulmani⁷.

- **Scoring Direction:** Unscored (Triggers Family Alignment Guidance in Report)⁷.
- **Potential Ambiguity:** None.
- **Cultural/Context Concerns:** Essential for identifying cases where student interests conflict with parental expectations⁵.
- **Reading Difficulty Estimate:** FKGL 5.0
- **Class 10 Suitability:** Fully suitable.
- **Pilot Requirement:** Include in alpha pilot.

Item CTX-03: Long-Term Educational Aspiration Level

- **Construct Measured:** Educational Horizon & Degree Aspiration (Unscored Context)
- **Question:** Thinking about your education after Intermediate (10+2), what is your current long-term educational plan?
- **Answer Choices:**
 - (A) Complete a 3-year or 4-year professional Bachelor's degree (B.Tech, MBBS, B.Com, B.A., etc.)
 - (B) Complete a short 2-year or 3-year diploma or ITI course to start earning early
 - (C) Pursue advanced postgraduate study or professional degrees (Masters, CA, Research)
 - (D) I am still exploring options and do not know yet
- **Question Type:** Single-select categorical
- **Why It Measures the Construct:** Contextualizes educational timelines and financial horizons, useful for first-generation college aspirants¹.
- **Scoring Direction:** Unscored (Filters College Discovery Information)¹.
- **Potential Ambiguity:** None.
- **Cultural/Context Concerns:** Accommodates students balancing immediate livelihood needs against multi-year degree pathways¹.
- **Reading Difficulty Estimate:** FKGL 5.2
- **Class 10 Suitability:** Fully suitable.
- **Pilot Requirement:** Include in alpha pilot.

Item CTX-04: Preferred Everyday Work Setting

- **Construct Measured:** Physical Workplace Preference (Unscored Context)
- **Question:** When you imagine yourself working in the future, what type of everyday setting sounds most comfortable to you?
- **Answer Choices:**
 - (A) Working at an indoor desk using a computer and digital tools
 - (B) Working actively in outdoor fields, construction sites, or nature
 - (C) Working in laboratories, hospitals, or diagnostic testing centers
 - (D) Moving around to interact with people in offices, markets, or schools
- **Question Type:** Single-select categorical
- **Why It Measures the Construct:** Evaluates physical environmental preferences without biasing core vocational construct scores⁹.
- **Scoring Direction:** Unscored (Used for Qualitative Report Matching).
- **Potential Ambiguity:** None.

- **Cultural/Context Concerns:** Relatable across common occupational environments.
- **Reading Difficulty Estimate:** FKGL 4.7
- **Class 10 Suitability:** Fully suitable.
- **Pilot Requirement:** Include in alpha pilot.

Question Quality Review

A psychometric review of the 64 candidate items categorizes prompts by measurement strength, flags potential ambiguities, and identifies items requiring targeted pilot testing⁹.

Identified Strongest Items

Items characterized by high construct specificity, behavioral concreteness, socioeconomic neutrality, and strong curricular alignment:

- **ISI-01** (*Microscope/pond organisms*) and **ISI-04** (*Soil/plant testing*): Direct behavioral markers of biological curiosity and variable isolation that do not require expensive household equipment⁹.
- **QCR-02** (*Word problems into equations*) and **QCR-03** (*Step-by-step algorithms*): Isolate symbolic mathematical modeling and algorithmic thinking from routine arithmetic⁹.
- **TMD-01** (*Taking apart broken objects*) and **TMD-03** (*Electrical circuit building*): Measure mechanical and electrical hardware engagement through concrete, universal objects⁹.
- **CEE-01** (*Profit/discount calculations*) and **CEE-02** (*Market price dynamics*): Concrete commercial and economic scenarios directly connected to Class 10 Social Studies and Mathematics curricula³.
- **SHC-01** (*Peer teaching*) and **SHC-03** (*Civic welfare access*): Clear operationalization of human service and civic governance, key for CEC track alignment³.
- **CEA-01** (*Imaginative writing*) and **CEA-03** (*Card/banner layout design*): Assess verbal and visual aesthetic creativity through routine secondary school tasks⁴².

Items Requiring Targeted Pilot Testing

Items flagged for potential ceiling effects, gender skew, or differential item functioning (DIF) across language/region⁶:

- **ISI-05** (*Space and black holes*): At risk for high baseline endorsement (ceiling effect) due to general popular science media exposure. The pilot must evaluate whether this item discriminates between high and average science interest¹.
- **TMD-07** (*Automobile engines/gears*): Vocational interest literature shows significant gender-based mean differences on automotive items⁶. The pilot must verify whether it functions equitably across male and female students without distorting the overall TMD scale score⁶.
- **CEE-09** (*Comparing shopping app prices*): May demonstrate differential performance between urban students with personal smartphones and rural students with limited app access².
- **CEA-04** (*Acting on stage*): May correlate with extraversion rather than pure artistic/creative interest, potentially penalizing introverted creative individuals⁵.

Proposed Initial 40-Question Version

The operational 40-item battery consists of **36 scored interest items** (balanced at exactly 6 items per construct across the 6 dimensions) and **4 unscored contextual items**⁹. The scored items are ordered across four thematic sections to distribute constructs evenly and minimize response fatigue⁹.

Pos	Item ID	Construct	English Question Stem	Adapted Telugu Question Stem	Scoring Direction
01	CTX-01	Context	When studying for exams in Class 10, which subject area generally felt easiest and most natural for you to understand ?	10వ తరగతి పరీక్షలకు చదివేటప్పుడు, మీకు ఏ సబ్జెక్ట్ చాలా సులభంగా మరియు సహజంగా అర్థమైనట్లు అనిపించింది?	Unscored Filter
02	CTX-02	Context	What stream or career path do your parents or family elders most hope you will choose after Class 10?	10వ తరగతి తర్వాత మీరు ఏ గ్రూప్ లేదా కెరీర్ ఎంచుకోవాల్సినదని మీ తల్లిదండ్రులు లేదా పెద్దలు ఎక్కువగా కోరుకుంటున్నారు?	Unscored Filter
03	ISI-01	ISI	How much do you	మొక్కల ఆకులు,	Positive (

			enjoy using a microscope or magnifying lens to examine plant leaves, insects, or tiny pond water organisms?	కీటకాలు లేదా నీటిలోని సూక్ష్మజీవులను మైక్రోస్కోప్ లేదా భూతద్దం ద్వారా పరిశీలించడం మీకు ఎంతవరకు ఇష్టం?	1 → 5)
04	QCR-02	QCR	When faced with a word problem in math, how much do you enjoy translating it into algebraic equations to find the unknown value?	లెక్కల్లో ఒక సమస్యను సమీకరణాలుగా (equations) మార్చి పరిష్కారం కనుగొనడం మీకు ఎంతవరకు ఆసక్తికరంగా ఉంటుంది?	Positive (1 → 5)
05	TMD-03	TMD	How much do you enjoy connecting batteries, wires, small bulbs, and switches to build working electrical circuits?	బ్యాటరీలు, వైర్లు, చిన్న బల్బులు మరియు స్విచ్‌లను కలిపి ఎలక్ట్రికల్ సర్క్యూట్‌లను తయారుచేయడం మీకు ఎంతవరకు	Positive (1 → 5)

				ఇష్టం?	
06	CEE-01	CEE	How much do you enjoy calculating profit, loss, and discounts when buying items in bulk or planning a sale?	వస్తువులను కొనుగోలు చేసేటప్పుడు లేదా విక్రయించేటప్పుడు లాభం, నష్టం మరియు డిస్కౌంట్లను లెక్కించడం మీకు ఎంతవరకు ఇష్టం?	Positive (1 → 5)
07	SHC-01	SHC	How much do you enjoy patiently explaining a difficult homework topic to a classmate who is struggling to understand it?	అరం చేసుకోవడానికి ఇబ్బంది పడుతున్న మీ క్లాస్మేట్ కి ఒక కష్టమైన పాఠాన్ని ఓపికగా వివరించడం మీకు ఎంతవరకు ఇష్టం?	Positive (1 → 5)
08	CEA-01	CEA	How much do you enjoy writing imaginative poems,	మీ భావాలను వ్యక్తపరచడానికి కవితలు, కథలు లేదా	Positive (1 → 5)

			short stories, or personal diary entries to express your feelings?	డైరీ రాయడం మీకు ఎంతవరకు ఇష్టమైన పని?	
09	ISI-02	ISI	How much do you enjoy mixing household liquids (like lemon juice, baking soda, and vinegar) to observe chemical reactions and color changes?	నిమ్మరసం, వంటసోడా లాంటి పదార్థాల ను కలిపి రసాయన మార్పుల ను మరియు రంగుల మార్పుల ను గమనించడం మీకు ఎంతవరకు ఇష్టం?	Positive (1 → 5)
10	QCR-01	QCR	How much do you enjoy finding patterns in sequences of numbers or solving sudoku and logic number grids?	సంఖ్యల క్రమంలోని సూత్రాలను (patterns) కనుగొనడం లేదా సుడోకు మరియు లాజిక్ పజిల్స్ ను పరిష్కరించడం మీకు ఎంతవరకు ఇష్టం?	Positive (1 → 5)
11	TMD-02	TMD	How much	ఒక ఇల్లు,	Positive (

			do you enjoy drawing scale floor plans or diagrams of how a house, room, or bridge should be constructed?	గది లేదా వంతెన నిర్మాణానికి సంబంధించిన ప్లాన్లు లేదా డ్రాయింగ్ లు గీయడం మీకు ఎంతవరకు ఇష్టం?	1 → 5)
12	CEE-02	CEE	How much do you enjoy learning how local markets determine the daily prices of vegetables, petrol, gold, or grains?	కూరగాయలు, పెట్రోల్, బంగారం లేదా ధాన్యాల ధరలు మార్కెట్లో ఎలా నిర్ణయించబడతాయో తెలుసుకోవడం మీకు ఎంత ఆసక్తిగా ఉంటుంది?	Positive (1 → 5)
13	SHC-02	SHC	When friends have an argument or disagreement, how much do you enjoy stepping in to help	స్నేహితుల మధ్య గొడవ లేదా అభిప్రాయ భేదాలు వచ్చినప్పుడు, పరిష్కారం చూపి వారిని కలపడం	Positive (1 → 5)

			them find a peaceful, fair solution?	మీకు ఎంతవరకు ఇష్టం?	
14	CEA-02	CEA	How much do you enjoy sketching pencil portraits, landscape drawings, or digital poster illustrations ?	పెన్సిల్తో ముఖచిత్రాలు, ప్రకృతి దృశ్యాలు గీయడం లేదా పోస్టరు డిజైన్ చేయడం మీకు ఎంతవరకు ఇష్టం?	Positive (1 → 5)
15	ISI-04	ISI	How much do you enjoy tracking why a plant in your garden is drying up by testing different soil types and watering schedules?	తోటలోని మొక్క ఎండిపోతుంటే, నేల రకాన్ని మరియు నీటి పరిమాణాన్ని మారుస్తూ పరిశీలించడం మీకు ఎంతవరకు ఇష్టం?	Positive (1 → 5)
16	QCR-03	QCR	How much do you enjoy writing down step-by-step instructions	ఏదైనా ఒక చిక్కు ప్రశ్నను ఇతరులు సులభంగా పరిష్కరించేలా దశలవారీ	Positive (1 → 5)

			(algorithms) so someone else can solve a complex riddle without making mistakes?	గా నియమాలను రూపొందించడం మీకు ఎంతవరకు ఇష్టం?	
17	TMD-01	TMD	How much do you enjoy taking apart a broken mechanical object (like a bicycle bell, toy, or pen) to see how its gears and springs work?	పాడైపోయిన సైకిల్ బెల్, బొమ్మ లేదా పెన్ను విప్పి, అందులోని స్ప్రింగ్లు మరియు భాగాలు ఎలా పనిచేస్తున్నాయో చూడటం మీకు ఎంతవరకు ఇష్టం?	Positive (1 → 5)
18	CEE-04	CEE	How much do you enjoy maintaining a neat, accurate logbook of income, pocket money, and expenses to track your savings?	మీ ఖర్చులు మరియు పొదుపును లెక్కించడానికి పాకెట్ మనీ వివరాలను ఒక పుస్తకంలో క్రమపద్ధతిలో నమోదు చేయడం	Positive (1 → 5)

				మీకు ఎంతవరకు ఇష్టం?	
19	SHC-03	SHC	How much do you enjoy learning about government welfare programs and how citizens can access their legal rights and benefits?	ప్రభుత్వ సంక్షేమ పథకాలు మరియు ప్రజలకు లభించే చట్టపరమైన హక్కుల గురించి తెలుసుకోవడం మీకు ఎంతవరకు ఆసక్తికరం?	Positive (1 → 5)
20	CEA-03	CEA	How much do you enjoy selecting colors, fonts, and photo layouts to make an invitation card or banner look attractive?	శుభలేఖలు లేదా బ్యానర్లు అందంగా కనిపించేలా రంగులు, అక్షరాలు మరియు ఫోటోలను ఎంపిక చేయడం మీకు ఎంతవరకు ఇష్టం?	Positive (1 → 5)
21	QCR-04	QCR	How much do you enjoy calculating speed, time, and	రెండు ఊర్ల మధ్య త్వరగా చేరుకోవడానికి వేగం, సమయం	Positive (1 → 5)

			distance to figure out the fastest travel route between two towns?	మరియు దూరాన్ని లెక్కించడం మీకు ఎంతవరకు ఇష్టం?	
22	TMD-04	TMD	How much do you enjoy using hand tools (like screwdrivers, pliers, or hammers) to repair a loose hinge or fix furniture at home?	ఇంట్లో స్క్రూడ్రైవర్, ప్లయిర్స్ వంటి పరికరాలను ఉపయోగించి వస్తువులను బాగుచేయడం మీకు ఎంతవరకు ఇష్టం?	Positive (1 → 5)
23	CEE-03	CEE	How much do you enjoy planning how to advertise and pitch a new product to convince classmates to buy it for a school fundraiser?	పాఠశాల కార్యక్రమంలో ఏదైనా వస్తువును ఇతరులు కొనేలా ఆకట్టుకునే మాటలతో ప్రచారం చేయడం మీకు ఎంతవరకు ఇష్టం?	Positive (1 → 5)
24	SHC-04	SHC	How much do you enjoy volunteerin	రకదాన శిబిరాలు, ఉచిత వైద్య	Positive (1 → 5)

			g your time to help organize health camps, blood donation drives, or neighborhood cleanups?	శిబిరాలు లేదా పరిసరాల పరిశుభ్రత కార్యక్రమాల్లో వాలంటీర్ గా పాల్గొనడం మీకు ఎంతవరకు ఇష్టం?	
25	CEA-05	CEA	How much do you enjoy analyzing movies or web series to figure out how camera angles, background music, and storytelling create emotion?	సినిమాలో కథ, నేపథ్య సంగీతం మరియు కెమెరా దృశ్యాలు ప్రేక్షకులను ఎలా ఆకట్టుకుంటున్నాయో విశ్లేషించడం మీకు ఎంతవరకు ఇష్టం?	Positive (1 → 5)
26	ISI-03	ISI	How much do you enjoy reading articles or watching videos about how human vaccines, medicines, and the	వ్యాక్సిన్లు, మందులు మరియు శరీర రోగనిరోధక వ్యవస్థ వ్యాధులతో ఎలా పోరాడుతాయో తెలుసుకోవడం మీకు	Positive (1 → 5)

			immune system fight infections?	ఎంతవరకు ఆసక్తికరం?	
27	QCR-05	QCR	How much do you enjoy organizing scattered facts or exam marks into neat data tables, bar charts, and pie graphs?	వివరాలు లేదా మార్కుల పట్టికలను నిశ్చితంగా పరిశీలించి బార్ చార్టులు మరియు గ్రాఫ్‌లుగా అమర్చడం మీకు ఎంతవరకు ఇష్టం?	Positive (1 → 5)
28	TMD-05	TMD	How much do you enjoy reading instruction manuals or diagrams that explain how a new electronic machine works?	కొత్త ఎలక్ట్రానిక్ వస్తువు ఎలా పనిచేస్తుంది దో తెలిపే మాన్యువల్ లేదా రేఖాచిత్రాలను చదవడం మీకు ఎంతవరకు ఇష్టం?	Positive (1 → 5)
29	CEE-05	CEE	How much do you enjoy reading or discussing news	భారతదేశంలో కొత్త వ్యాపారాలు (స్టార్టప్‌లు), బ్యాంకులు	Positive (1 → 5)

			stories about new business startups, banks, and economic growth in India?	మరియు ఆర్థిక అభివృద్ధికి సంబంధించిన వార్తలను తెలుసుకోవడం మీకు ఎంత ఆసక్తికరం?	
30	SHC-05	SHC	When someone in your family or circle feels sad, how much do you enjoy listening to their personal worries and comforting them?	కుటుంబంలో లేదా స్నేహితుల్లో ఎవరైనా బాధలో ఉన్నప్పుడు, వారి సమస్యలను వినే ఓదార్చడం మీకు ఎంతవరకు సహజంగా అనిపిస్తుంది?	Positive (1 → 5)
31	CEA-04	CEA	How much do you enjoy performing in school plays, acting out characters, or reciting dramatic dialogues on stage?	నాటకాల్లో నటించడం, విభిన్న పాత్రలు వేయడం లేదా స్టేజ్ పై డైలాగులు చెప్పడం మీకు ఎంతవరకు ఇష్టం?	Positive (1 → 5)
32	ISI-06	ISI	How much	పువ్వు	Positive (

			do you enjoy dissecting flower petals and seeds to understand their inner reproductive parts?	భాగాలు మరియు వితనాల లోపేలి నిర్మాణాన్ని తెలుసుకోవడానికి వాటిని విడదీసి చూడటం మీకు ఎంతవరకు ఆసక్తికరం?	1 → 5)
33	QCR-06	QCR	How much do you enjoy calculating probability, such as figuring out the exact chances of winning a game with dice or coins?	పాచికలు లేదా కాయిన్స్ ఆటలో గెలిచే అవకాశాలను లెక్కించడం మీకు ఎంతవరకు ఆసక్తిగా ఉంటుంది?	Positive (1 → 5)
34	TMD-06	TMD	How much do you enjoy building sturdy, three-dimensional models (like towers, vehicles, or cardboard bridges) using basic	అట్టపెట్టెలు మరియు సాధారణ వస్తువులతో వంతెనలు, భవనాలు లేదా వాహనాల నమూనాలను నిర్మించడం మీకు	Positive (1 → 5)

			craft materials?	ఎంతవరకు ఇష్టం?	
35	CEE-06	CEE	How much do you enjoy negotiating prices or terms with someone to reach a deal where both sides feel satisfied?	ఇరువైపులా న్యాయంగా ఉండేలా చర్చించి ఒప్పందం చేసుకోవడం మీకు ఎంతవరకు ఇష్టమైన విషయం?	Positive (1 → 5)
36	SHC-06	SHC	How much do you enjoy learning how a district court or village panchayat hears evidence to make fair legal judgments?	కోర్టులు లేదా గ్రామ పంచాయతీలు సాక్షులను పరిశీలించి తీర్పులు ఎలా ఇస్తాయో తెలుసుకోవడం మీకు ఎంత ఆసక్తికరం?	Positive (1 → 5)
37	CEA-06	CEA	How much do you enjoy practicing musical instruments , humming melodies, or arranging rhythms	సంగీత వాయిద్యాలను వాయింపడం, పాటలు పాడటం లేదా కొత్త ట్యూన్లు సృష్టించడం మీకు	Positive (1 → 5)

			and beats?	ఎంతవరకు ఇష్టం?	
38	ISI-08	ISI	How much do you enjoy learning the biological reasons behind why animals behave differently during seasonal migrations or weather changes?	వాతావరణ మార్పుల పుష్ట జంతువులు చేసే ప్రయాణాలు (వలసలు) మరియు వాటి ప్రవర్తనల వెనుక ఉన్న కారణాలను తెలుసుకోవడం మీకు ఎంత ఆసక్తికరం?	Positive (1 → 5)
39	CTX-03	Context	Thinking about your education after Intermediate (10+2), what is your current long-term educational plan?	ఇంటర్మీడియట్ (10+2) తర్వాత మీ పైచదువుల గురించి మీ ప్రస్తుత ఆలోచన ఏమిటి?	Unscored Filter
40	CTX-04	Context	When you imagine yourself working in the future, what type of everyday setting	భవిష్యత్తులో మీరు పని చేసే వాతావరణం ఎలా ఉండాలని కోరుకుంటున్నారు?	Unscored Filter

			sounds most comfortabl e to you?		
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Scoring Framework Proposal

The scoring system aggregates raw Likert responses into standardized construct scores and computes composite stream alignment indices without relying on uncalibrated weights⁹.

Step 1: Raw Construct Aggregation

For each construct dimension $k \in \{\text{ISI, QCR, TMD, CEE, SHC, CEA}\}$, the raw score R_k is the sum of responses across its 6 assigned items⁹:

$$R_k = \sum_{i=1}^6 x_{k,i}$$

Because responses range from 1 to 5, the raw score range is:

$$6 \leq R_k \leq 30$$

Step 2: Scale Standardization (POMP and T -Scores)

During the initial pre-normative operational window, scale scores are expressed as Percent of Maximum Possible (POMP) scores to provide an intuitive 0–100 scale:

$$\text{POMP}_k = \left(\frac{R_k - 6}{24} \right) \times 100$$

Following large-scale normative pilot testing, raw scores are converted into standardized T -scores based on empirical population parameters (μ_k and σ_k) to eliminate scale elevation differences⁹:

$$T_k = 50 + 10 \left(\frac{R_k - \mu_k}{\sigma_k} \right)$$

Step 3: Stream Alignment Composite Indices

Stream alignment scores are calculated as linear combinations of standardized construct scores (T_k), reflecting the subject demands of BIEAP and TSBIE curricula³:

$$\text{Score}_{\text{MPC}} = 0.50(T_{\text{QCR}}) + 0.35(T_{\text{TMD}}) + 0.15(T_{\text{ISI}})$$

$$\text{Score}_{\text{BiPC}} = 0.60(T_{\text{ISI}}) + 0.25(T_{\text{SHC}}) + 0.15(T_{\text{QCR}})$$

$$\text{Score}_{\text{MEC}} = 0.45(T_{\text{CEE}}) + 0.40(T_{\text{QCR}}) + 0.15(T_{\text{SHC}})$$

$$\text{Score}_{\text{CEC}} = 0.50(T_{\text{CEE}}) + 0.35(T_{\text{SHC}}) + 0.15(T_{\text{CEA}})$$

Assumptions Requiring Empirical Validation

The relative weights in the stream alignment formulas represent operational hypotheses based on curriculum subject distributions, not verified structural coefficients³. These weights carry three primary assumptions that must be empirically evaluated during beta testing:

- **Curricular Proportionality Assumption:** The model assumes that student interest in MPC is driven primarily by Quantitative Reasoning (50%) and Mechanical/Applied Thinking (35%)³. If longitudinal validation reveals that Chemistry demands in Intermediate require higher scientific inquiry, the ISI weight must be adjusted upward⁴.
- **Linear Additivity Assumption:** The formulas treat construct interests as compensatory (e.g., high quantitative interest compensating for low mechanical interest in MPC). Research must investigate whether non-linear conjunctive thresholds (requiring minimum levels across both QCR and TMD) better predict track persistence⁵.
- **Weights Calibration Protocol:** The proposed weights remain provisional until large-scale field testing (Phase 5). At that point, structural equation modeling (SEM) and multiple regression will derive empirical weights by regressing student stream satisfaction and academic persistence onto construct scores⁹.

Pilot/Validation Plan

To establish empirical validity while working within early-stage operational constraints, Naaguru will implement an eight-phase validation plan adhering to ITC standards¹⁰.

Phase	Target Sample Size & Cohort	Primary Operational Objectives	Analytical Metrics & Criteria
1. Expert Review	$N =$ Subject-Matter Experts (2)	Evaluate content validity, construct coverage, developmental	Content Validity Index ($CVI \geq$)

	Psychologists, 2 Senior Secondary Teachers, 1 Telugu Linguist)	tone, and translation fidelity ¹⁶ .	across all items ⁶⁰ .
2. Cognitive Debriefing	$N =$ Class 10 Students (Stratified by gender, language medium, and location: Hyderabad, Guntur, Mahbubnagar)	Conduct think-aloud cognitive interviews to observe item processing, identify awkward phrasing, and record completion times ¹⁶ .	Qualitative item revision where $>$ of students misinterpret item intent ⁵⁴ .
3. Alpha Pilot	$N =$ Class 10 Students across 4 partner schools (2 government, 2 private)	Administer full 64-item candidate pool to examine response variance and preliminary scale cohesion ⁹ .	Screen for item skewness (\$
4. Bank Refinement	Psychometric Research Team	Down-select 64-item pool to operational 40-item battery (36 scored + 4 context items) based on alpha pilot metrics ⁹ .	Select top 6 psychometrically functioning items per construct scale ⁹ .
5. Beta Field Pilot	$N =$ Class 10 Students (Stratified across government ZP schools and private junior colleges)	Large-scale standardization to establish regional norms (μ, σ), evaluate factor structure, and assess reliability ⁹ .	• Scale Cronbach's $\alpha \geq$ • McDonald's $\omega \geq$ • Confirmatory Factor Analysis: $CFI \geq$, $RMSEA \leq$ ⁹.

6. Cross-Language DIF Analysis	Bilingual Sub-sample ($N =$ English, $N =$ Telugu)	Differential Item Functioning (DIF) analysis to ensure items function equivalently across languages ¹⁰ .	Logistic regression / Mantel-Haenszel DIF: items with $R^2\Delta >$ flagged for re-translation ¹⁰ .
7. Test-Retest Reliability	$N =$ Students re-tested across a 14-day interval	Assess temporal measurement stability across the two-week decision window ⁹ .	Intraclass Correlation Coefficient ($ICC \geq$) and Pearson $r \geq$ ⁹ .
8. Longitudinal Tracking	$N =$ Students tracked into Intermediate 1st Year	Measure predictive criterion validity by evaluating correlation between predicted stream fit and 1st-year subject satisfaction ⁹ .	Point-biserial correlation $r \geq$ predicting academic satisfaction and stream persistence ⁵ .

Risks and Things Naaguru Must NOT Claim

To maintain institutional integrity, adhere to consumer protection norms, and safeguard students from educational misdirection, Naaguru must uphold strict operational boundaries⁵.

Explicitly Forbidden Claims

- **No Claim of Measuring Cognitive Aptitude or IQ:** The assessment does not evaluate intellectual capability, general mental ability (g), or innate academic talent⁸. Statements suggesting that test scores prove a student is "naturally gifted at mathematics" or "lacks scientific intelligence" are strictly prohibited⁸.
- **No Deterministic Career Predictions:** The platform must never claim that a student is "destined" or "meant" to pursue a specific occupation⁵. The platform must never guarantee that following a recommendation will lead to employment, entrance examination success, or specific earnings⁸.
- **No Diagnostic Clinical Profiling:** The assessment does not evaluate mental health, psychological adjustment, emotional stability, or personality disorders²⁴.
- **No Guarantee of Intermediate Board Exam or Competitive Entrance Rank:** High scores in quantitative interest do not guarantee passing or ranking high in IIT-JEE, NEET,

or state EAPCET examinations³.

Operational Ethical Standards

- **Constructive Family Engagement:** Results must be delivered with educational explanations designed to support constructive family discussions, helping alleviate the familial pressure highlighted by Arulmani's career research⁷.
- **Student Data Protection:** Assessment data collected from minors aged 15 to 17 must comply with India's Digital Personal Data Protection Act (DPDPA). Student response records must never be sold or transferred to commercial coaching institutes or private college marketing desks without verified parental consent.
- **Continuous Bias Monitoring:** If ongoing pilot tracking indicates that specific items display systematic gender, caste, or regional bias, those items must be adjusted or replaced to ensure fair, equitable educational guidance for all students⁶.

Source Bibliography

This section synthesizes the empirical evidence, official educational frameworks, and psychometric standards supporting the assessment design, categorizing each design decision by its evidentiary basis³.

Source Citation & Academic Identifier	Authors / Sponsoring Organization	Year	Operational Link / Reference	Specific Supported Findings & Operational Claims	Classification of Assessment Choice
<i>Normative Changes in Interests from Adolescence to Adulthood: A Meta-Analysis of Longitudinal Studies</i> [cite: 20]	Su, R., Stoll, G., & Rounds, J. (Psychological Bulletin)	2018	<u>DOI: 10.1037/bul0000165</u>	Demonstrates that vocational interests exhibit meaningful developmental changes across adolescence, supporting the use of age-adapted	RESEARCH EVIDENCE

				typical-performance interest measurement rather than static adult inventories ²⁰ .	
<i>The Stability of Vocational Interests From Early Adolescence to Middle Adulthood: A Quantitative Review of Longitudinal Studies</i> [cite: 21]	Low, K. S. D., Yoon, M., Roberts, B. W., & Rounds, J. (Psychological Bulletin)	2005	DOI: 10.1037/0033-2909.131.5.713	Confirms that the rank-order stability of vocational interests is already substantive by age 15 ($r \approx$), establishing that measuring interest profiles at Class 10 is psychometrically reliable for guidance ²¹ .	RESEARCH EVIDENCE
<i>Cross-Cultural Structural Equivalence of RIASEC Models and Measures</i> [cite: 14]	Rounds, J., & Tracey, T. J. (Journal of Counseling Psychology)	1996	DOI: 10.1037/0022-0167.43.3.310	Establishes via structural meta-analysis that Holland's circular order (hexagonal	RESEARCH EVIDENCE

				calculus) fails to fit non-Western samples, demonstrating that direct importation of Western inventories without structural adaptation is invalid ¹⁴ .	
<i>The Cultural Preparation Process Model and Career Development</i> [cite: 7]	Arulmani, G. (Handbook of Career Development: International Perspectives)	2014	<u>Springer Publishing, pp. 609–630</u>	Defines the socio-cognitive environment of Indian career choices, demonstrating the dominant role of family expectations, occupational prestige hierarchies, and the need to decouple task interest from social status ⁷ .	RESEARCH EVIDENCE
<i>Assessment of Interest</i>	Arulmani, G.	2014	<u>Springer Publishing,</u>	Delineates the	RESEARCH EVIDENCE

<i>and Aptitude: A Methodologically-Integrated Approach</i> [cite: 8]	(Handbook of Career Development)		<u>pp. 609–630</u>	theoretical and empirical divide between interest and aptitude in Indian adolescents , warning that self-reported interest in science often contradicts actual cognitive readiness⁸.	
<i>ITC Guidelines for Translating and Adapting Tests (Second Edition)</i> [cite: 10]	International Test Commission (International Journal of Testing)	2017	<u>DOI: 10.1080/15305058.2017.1398166</u>	Sets international standards for cross-lingual psychometric equivalence , mandating dual forward translation, expert reconciliation, blind back-translation, and empirical DIF verification¹⁰.	RESEARCH EVIDENCE

<i>O*NET Interest Profiler Short Form: Psychometric Properties and Adolescent Validity</i> [cite: 9, 33]	Rounds, J., Su, R., Lewis, P., & Rivkin, D. (National Center for O*NET Development)	2010	<u>O*NET Resource Center</u>	Provides the psychometric blueprint for brief 60-item activity-based interest profiling with validated scale reliabilities ($\alpha =$), supporting Naaguru's task-based item design ⁹ .	RESEARCH EVIDENCE
<i>Telangana State Board of Intermediate Education Exam Regulations and Curriculum Framework</i> [cite: 3]	Telangana State Board of Intermediate Education (TSBIE)	2026	<u>Official TSBIE Portal</u>	Establishes the official subject distributions, mark allocations, and curricular parameters for Intermediate MPC, BiPC, MEC, and CEC streams in Telangana ³ .	RESEARCH EVIDENCE
<i>Board of Intermediate Education</i>	Board of Intermediate	2025	<u>BIEAP Official Gazette</u>	Outlines syllabus restructuring	RESEARCH EVIDENCE

<i>Reforms & Curriculum Restructuring</i> [cite: 4]	Education, Andhra Pradesh (BIEAP)			g, subject combinations, and examination grading patterns across science and humanities tracks in Andhra Pradesh ⁴ .	
<i>How Accurate Are Interest Inventories? A Quantitative Review of Career Choice Hit Rates</i> [cite: 28]	Nye, C. D., Su, R., Rounds, J., & Drasgow, F. (Psychological Bulletin)	2021	<u>DOI: 10.1037/bul0000269</u>	Synthesizes meta-analytic evidence showing a 50.8% overall hit rate for interest inventories predicting occupational choice, supporting their use as probabilistic guidance tools rather than definitive predictors ²⁸ .	RESEARCH EVIDENCE
<i>Class 10 State Board Curriculum and Language Assessment Standards</i>	State Council of Educational Research and Training (SCERT AP &	2025	<u>SCERT Curriculum Documentation</u>	Provides baseline reading standards, textbook lexical ranges, and	RESEARCH EVIDENCE

[cite: 18, 39]	Telangana)			subject chapter content for Class 10 SSC students in Andhra Pradesh and Telangana ¹⁸ .	
Selection of 5-Point Likert Scale with Behavioral Anchors over Forced-Choice Formats	Naaguru Psychometric Architecture Team	2026	Internal Assessment Design Decision	Adopting a 5-point Likert scale to minimize mobile cognitive load and generate normative scores for stream mapping, addressing acquiescence bias through concrete behavioral anchors ¹⁹ .	PRODUCT DECISION
Operationalization of Exactly Six Non-Redundant Interest Constructs (ISI, QCR, TMD, CEE, SHC, CEA)	Naaguru Psychometric Architecture Team	2026	Internal Assessment Design Decision	Eliminating "Leadership" and broad "Analytical" categories, replacing them with six targeted dimensions that map directly	PRODUCT DECISION

				onto BIEAP/TSBIE intermediate streams ³ .	
Inclusion of 4 Unsourced Contextual Items (CTX-01 to CTX-04)	Naaguru Psychometric Architecture Team	2026	Internal Assessment Design Decision	Capturing family expectations, academic ease, and educational horizons to provide contextual filters in student reports without contaminating psychometric interest scores ⁷ .	PRODUCT DECISION
Target Readability Ceiling of FKGL 4.0 to 5.5 for Class 10 Items	Lead Assessment Researcher Inference	2026	Assessment Design Synthesis	Lowering reading difficulty relative to chronological grade level accommodates the functional English comprehension of regional-medium and semi-urban learners ² .	YOUR INFERENCE

Intermediate Stream Alignment Formulas and Relative Construct Weightings	Lead Assessment Researcher Inference	2026	Assessment Design Synthesis	Developing provisional linear composite weights based on board curriculum distributions (e.g., 50% QCR and 35% TMD for MPC), marked for empirical calibration during beta testing ³ .	YOUR INFERENCE
Contemporary Educational Telugu (Vyavaharika) Register Selection	Lead Assessment Researcher Inference	2026	Assessment Design Synthesis	Using contemporary spoken educational Telugu with common English loanwords avoids comprehension barriers caused by formal Sanskritized terms (<i>Grandhika</i>) ¹⁰ .	YOUR INFERENCE

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